

Nonlinear Characterization and Modeling Through Pulsed IV/S-Parameters

PRESENTERS



Hiroyuki Maehara



Tony Gasseling



Steve Dudkiewicz



Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

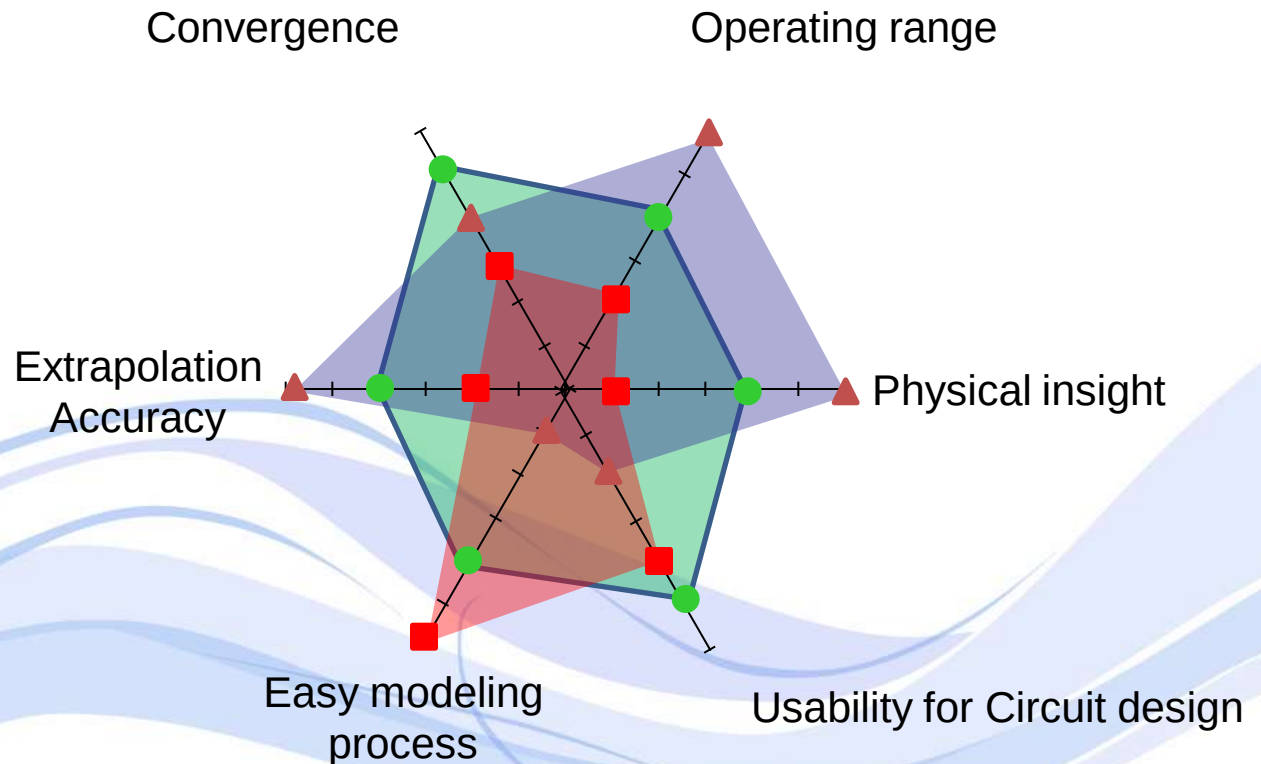
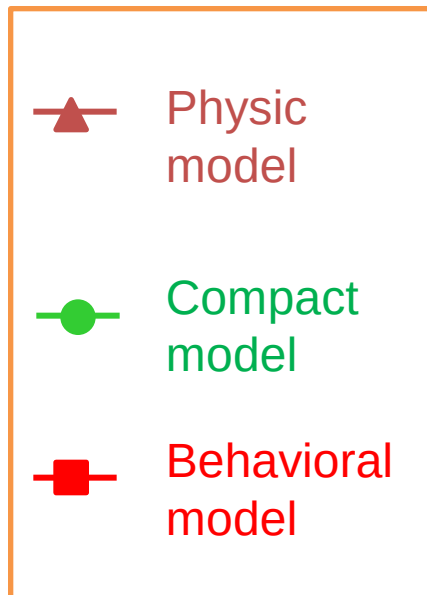
OUTLINE

- Introduction
- Core device model extraction
- Model Enhancement
- Model Validation

Your complete measurement & modeling solutions partner!



Types of Large-Signal Transistor Models





Commercial compact FET models

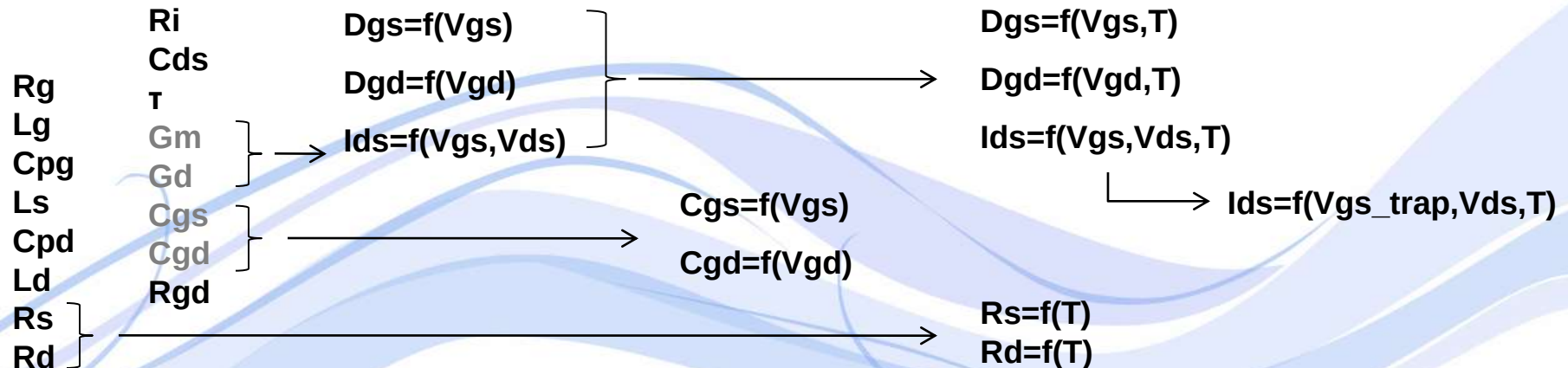
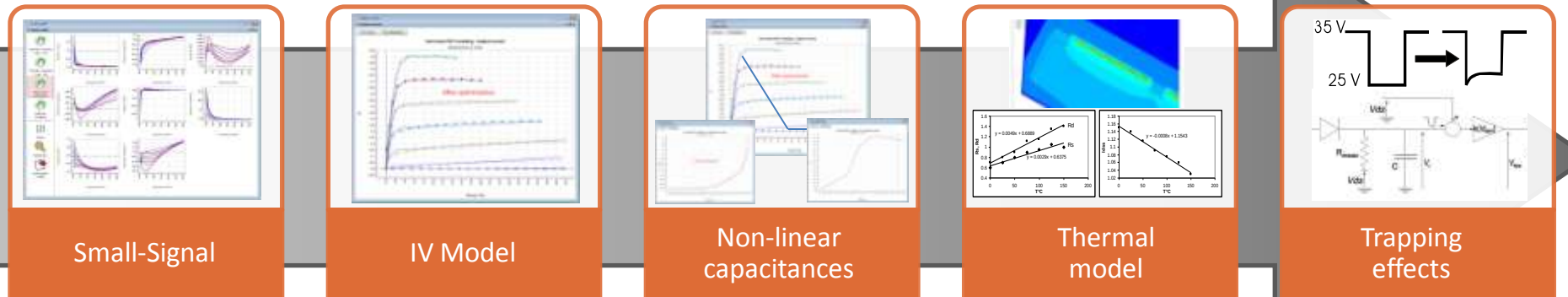
- Mostly used models for GaN HEMTs

FET models	Number of parameters	Electro-thermal effect	Trapping Effects	Original Device Context
Curtice3 [1]	59	No	No	GaAs FET
CFET [2]	53	Yes	No	HEMT
EEHEMT1 [3]	71	No	No	HEMT
Angelov [4]	80	Yes	No	HEMT/MESFET
AMCAD HEMT1 [5]	65	Yes	Yes	GaN HEMT

- AMCAD GaN HEMT1 is the only model here with a complete extraction flow based on pulsed IV/RF measurements



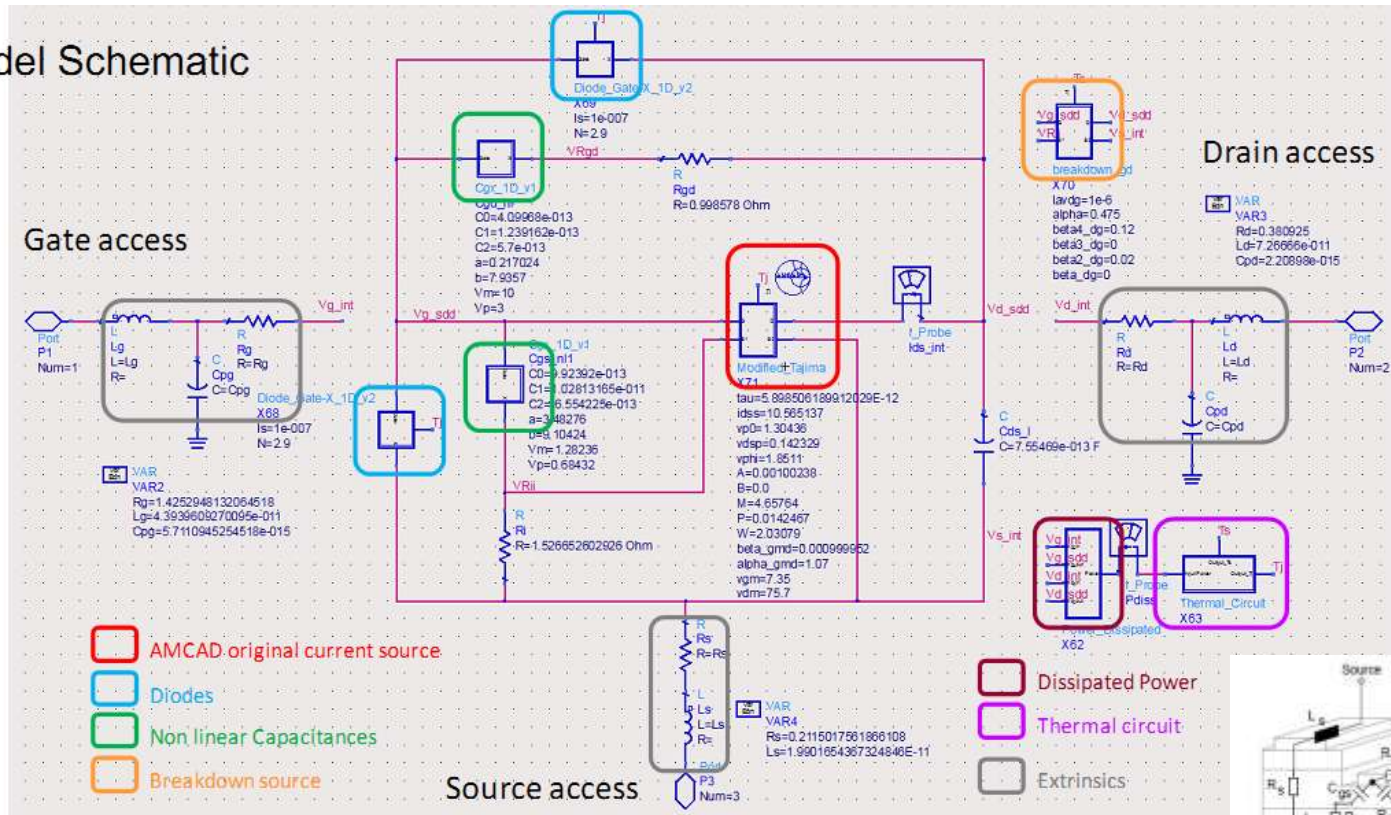
Compact FET model extraction flow



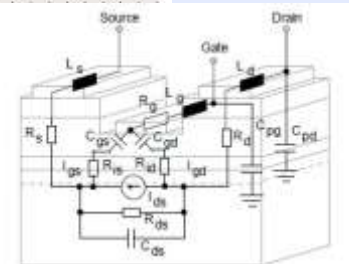
Various effects are successively added

Parameter extraction methodology

Model Schematic



- Based on 18-element AMCAD original current source
- Compatible with Agilent ADS





Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

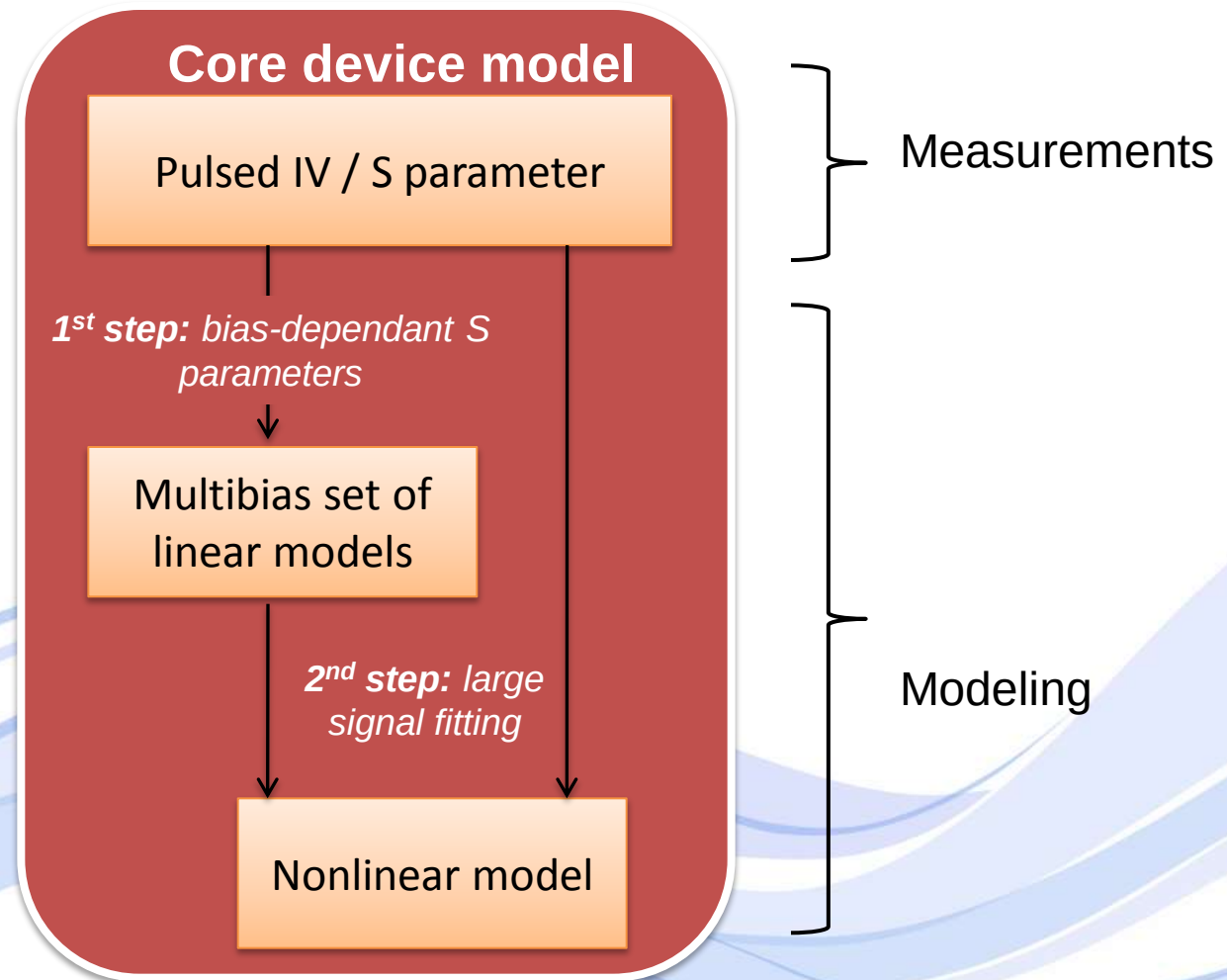
OUTLINE

- Introduction
- Core device model extraction
- Model Enhancement
- Model Validation

Your complete measurement & modeling solutions partner!

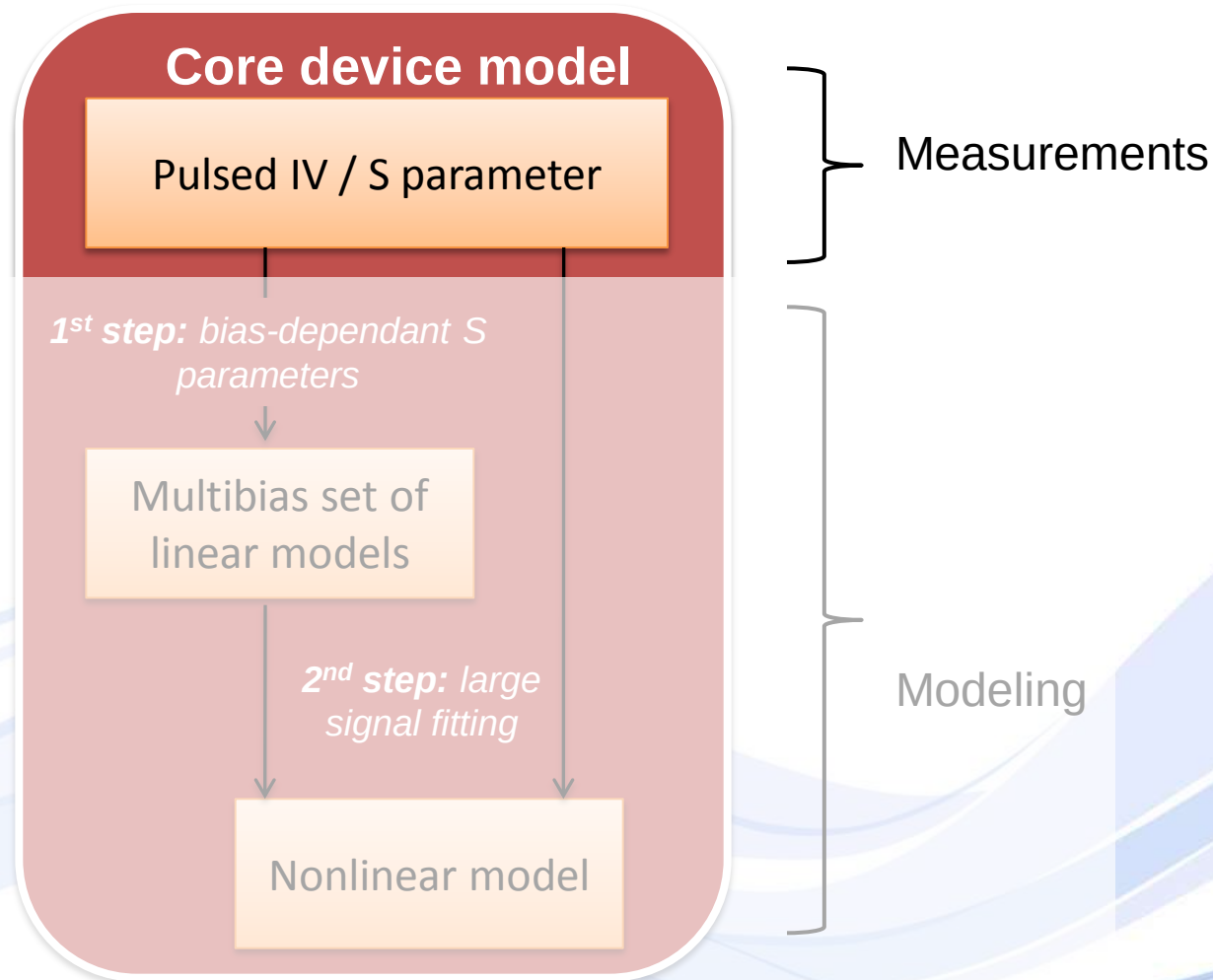


Parameter extraction methodology



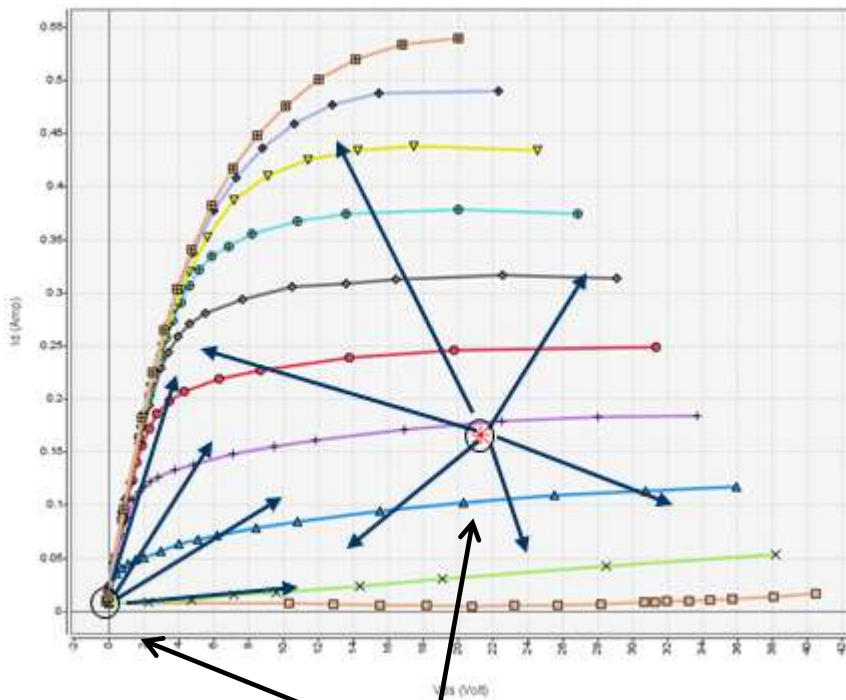


Parameter extraction methodology





Pulsed IV measurements

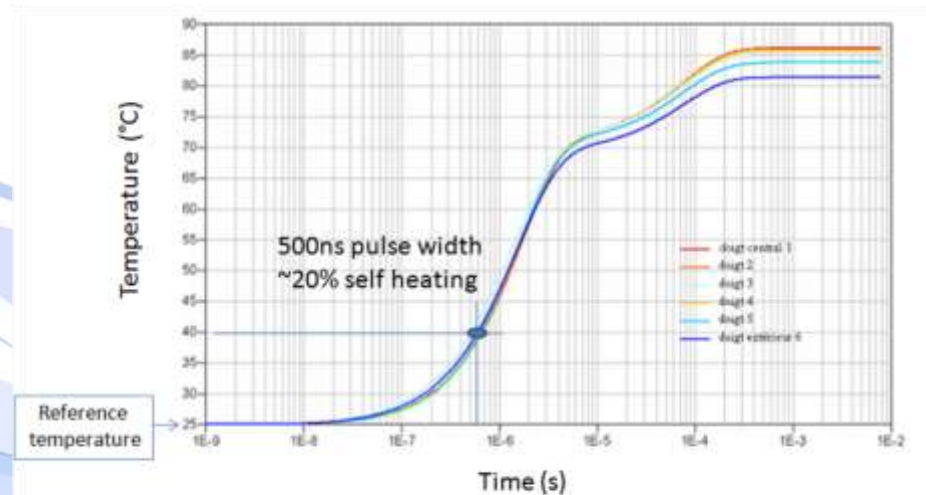


Several quiescent bias point

Short pulse : Quasi-isothermal conditions

Low duty cycle : Constant mean temperature

Quiescent bias point : Thermal conditions fixed



Pulsed Measurements = quasi isothermal operating conditions

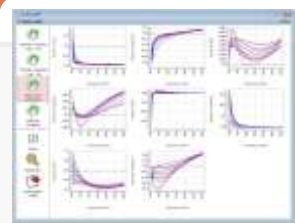


Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Compact FET model extraction flow



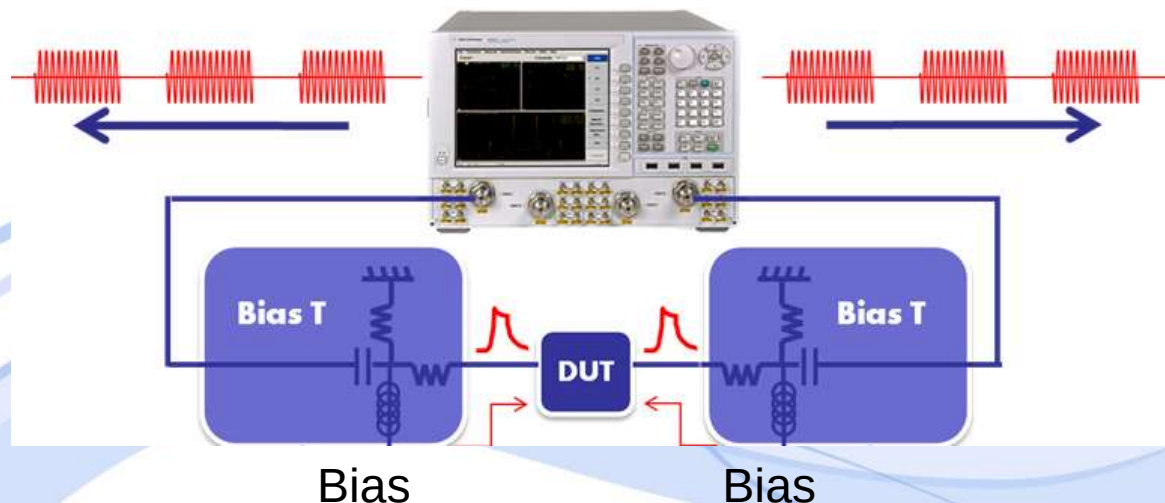
Small-Signal

Modelling Process

Pulsed S parameter measurements

Rg
Lg
Cpg
Ls
Cpd
Ld
Rs
Rd

Ri
Cds
 τ
Gm
Gd
Cgs
Cgd
Rgd



Your complete measurement & modeling solutions partner!

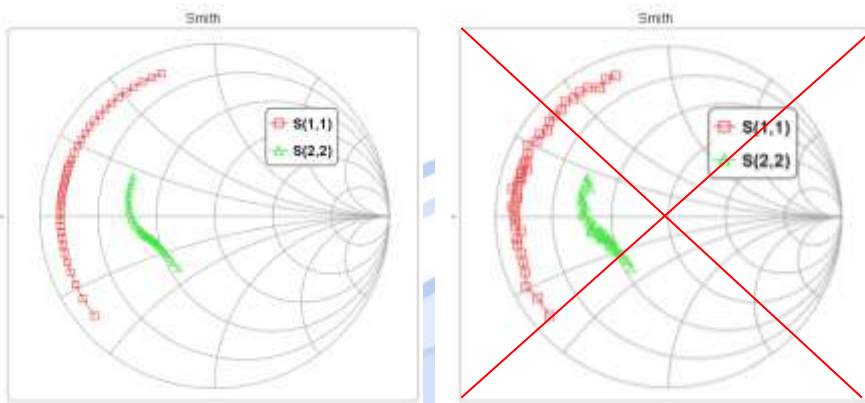


Pulsed S parameter measurements

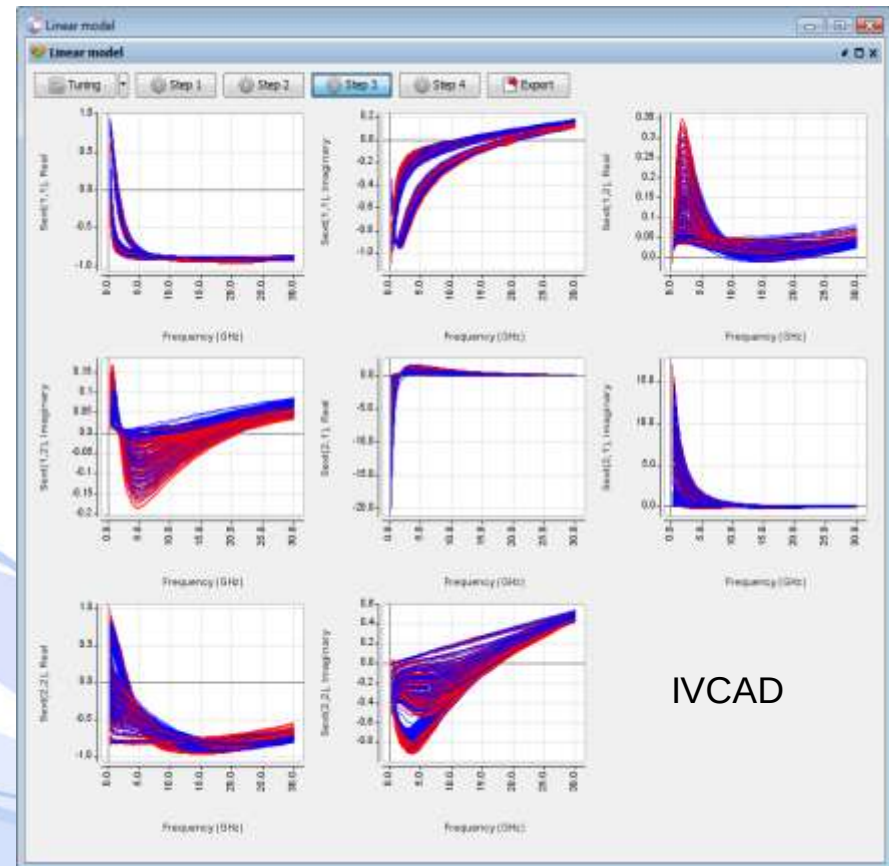
The first & most important point :

- Pulsed S parameter measurements must not be noisy
- Small S2P measurement variation = strong influence over the linear model extraction : optimization algorithm

Requirements :



Dynamic range in pulsed mode > 90dB for Duty Cycle ~ 5%



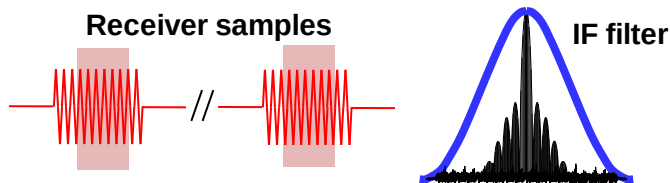


Pulsed S-parameter Measurements

Pulsed S-parameter measurements must not be noisy at low duty cycle with narrow pulse width

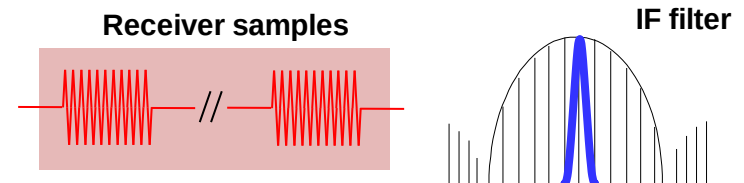
Pulse detection methods

Wideband detection



- No pulse desensitization
- Increased noise with narrow pulse width due to wider IF bandwidth
- Limited pulse width by maximum available IF bandwidth

Narrowband detection



- Narrower minimum pulse width than wideband pulse
- Reduced dynamic range with low duty cycle due to pulse desensitization by $20 \cdot \log(\text{duty cycle})$

PNA/PNA-X Noise reduction techniques and performances

Peak-to-peak noise with wideband detection at 10% duty cycle

No averaging in calibration and measurements



Pulse width (IFBW)

- 10 us (150 kHz)
- 5 us (280 kHz)
- 1 us (1.5 MHz)
- 500 ns (3 MHz)

Averaging 20 times in calibration and measurements

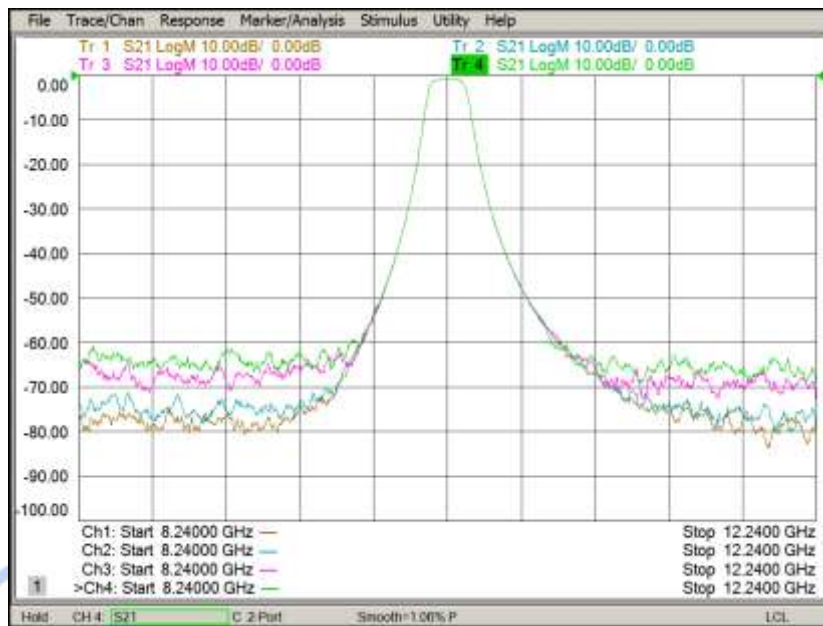




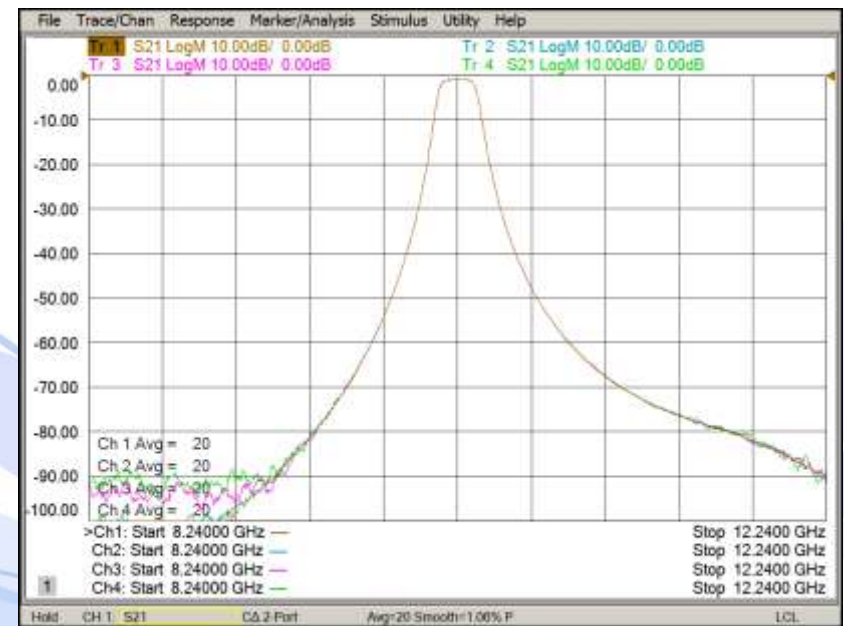
PNA/PNA-X Noise reduction techniques and performances

Dynamic range with wideband detection at 10% duty cycle with 10 us, 5 us, 1 us, 500 ns pulse width

No averaging in calibration and measurements, 1% smoothing on



Averaging 20 times in calibration and measurements, 1% smoothing on



PNA/PNA-X Noise reduction techniques and performances

Peak-to-peak noise at 10% duty cycle

Wideband detection with 20 times averaging in calibration and measurements



Pulse width

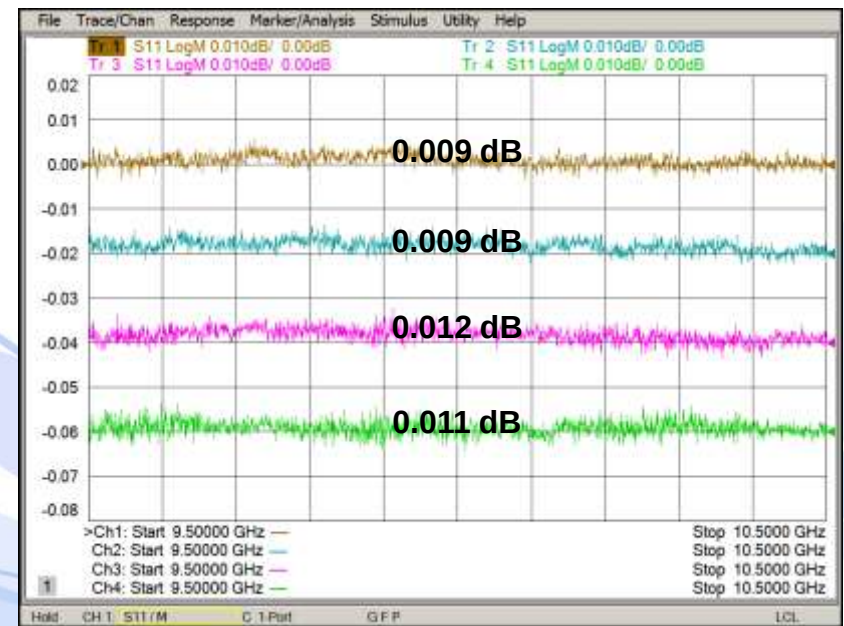
10 us

5 us

1 us

500 ns

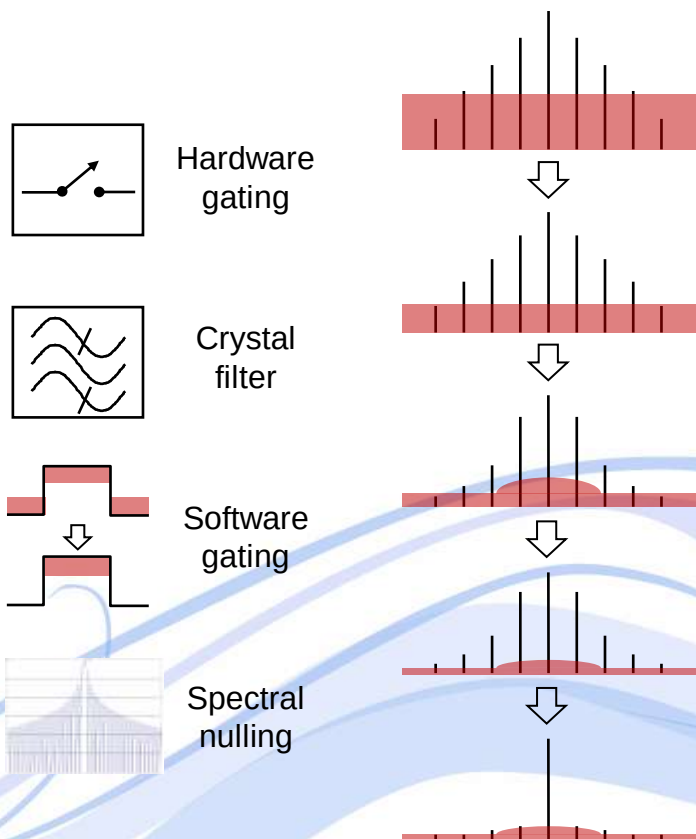
Narrowband detection with no averaging in calibration and measurements



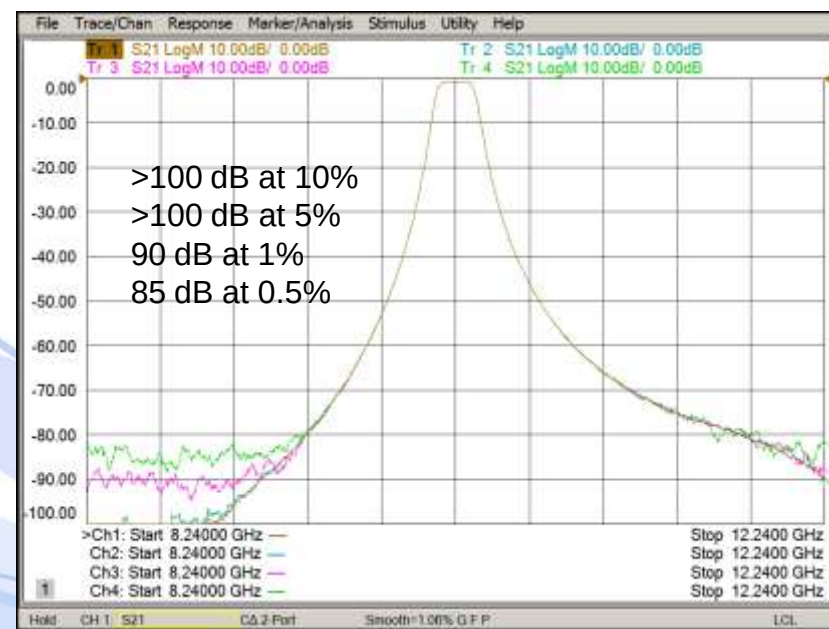


PNA/PNA-X Noise reduction techniques and performances

Dynamic range with narrowband detection at 500 ns pulse width



No Averaging, 1% smoothing on, 500 Hz IF bandwidth





	E836x Legacy PNA	N524xA PNA-X	N522xA New PNA
			
Pulse generator	External	Internal/External	Internal/External
Pulse modulator	External	Internal/External	Internal/External
Wideband detection			
Max BW/Min PW	35 kHz / 50 us	15 MHz / 100 ns	15 MHz / 100 ns
High level noise*	0.006 dBrms	0.002 dBrms	0.002 to 0.003 dBrms
Dynamic range**	114 to 123 dB	124 to 129 dB	127 dB
Narrowband detection			
Min IF gate width	20 ns	<20 ns	<20 ns
Dynamic range***	85 dB	<105 dB	<105 dB

* Specified as trace noise magnitude, at 20 GHz, at 1 kHz IF bandwidth

** Specified performance at 20 GHz, at 10 Hz IF bandwidth

*** Measured performance at 10 GHz at 10 Hz IF bandwidth, 1% duty cycle

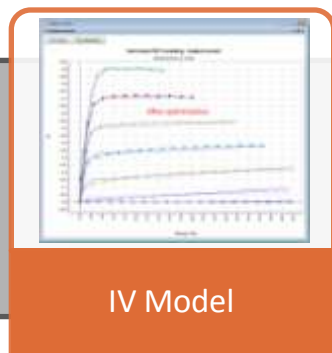
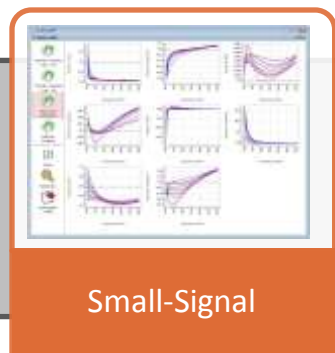


Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

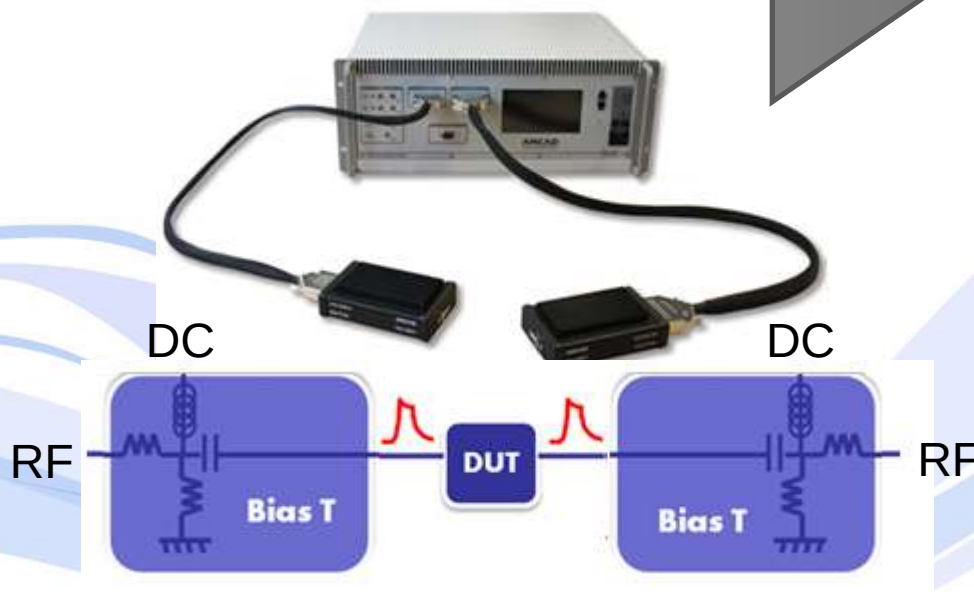
Compact FET model extraction flow



R_g
 L_g
 C_{pg}
 L_s
 C_{pd}
 L_d
 R_s
 R_d

R_i
 C_{ds}
 τ
 G_m
 G_d
 C_{gs}
 C_{gd}
 R_{gd}

$D_{gs}=f(V_{gs})$
 $D_{gd}=f(V_{gd})$
 $I_{ds}=f(V_{gs}, V_{ds})$

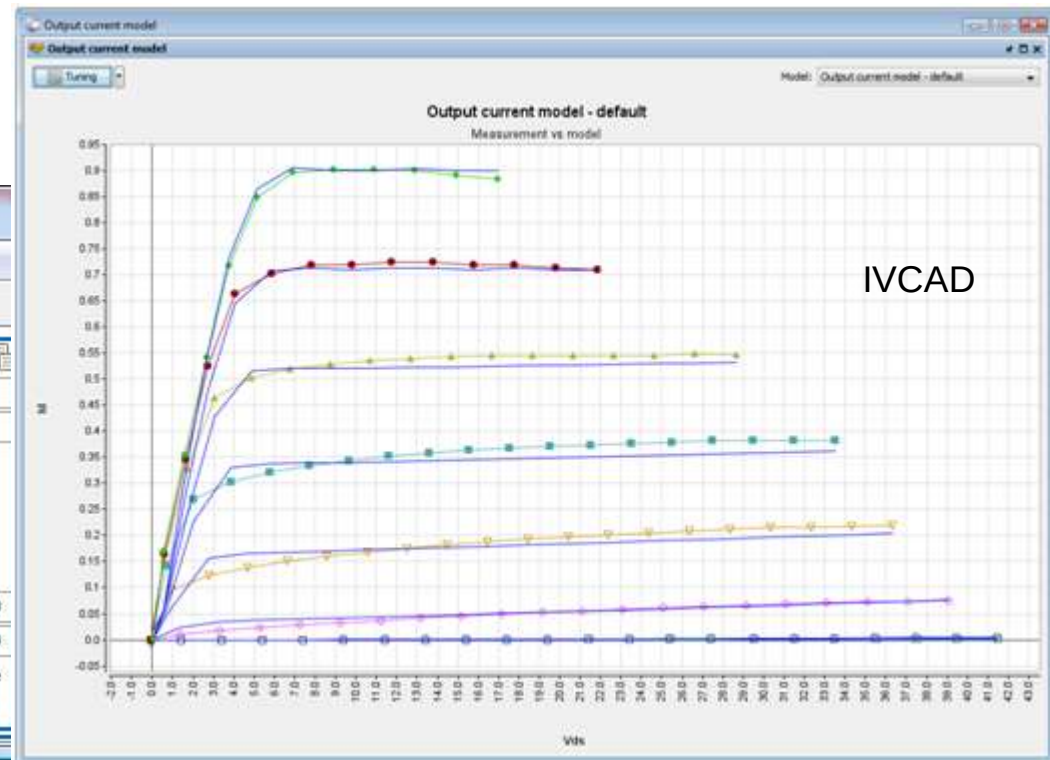
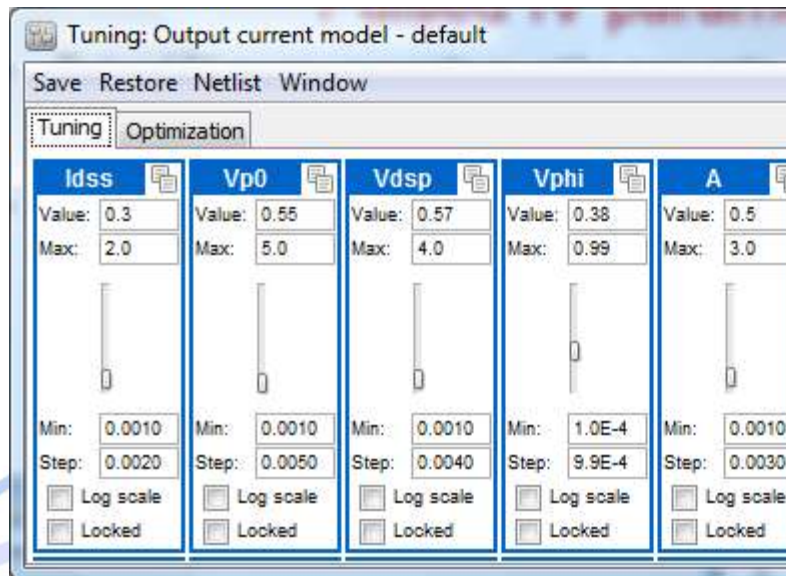


Your complete measurement & modeling solutions partner!



Pulsed IV parameter measurements

- Pulsed IV measurements must be accurate from low to high voltage/current values
- Accurate IV data =
 - Reliable current source
 - Transconductance
 - Leakage current
 - Ideality factor schottky diode

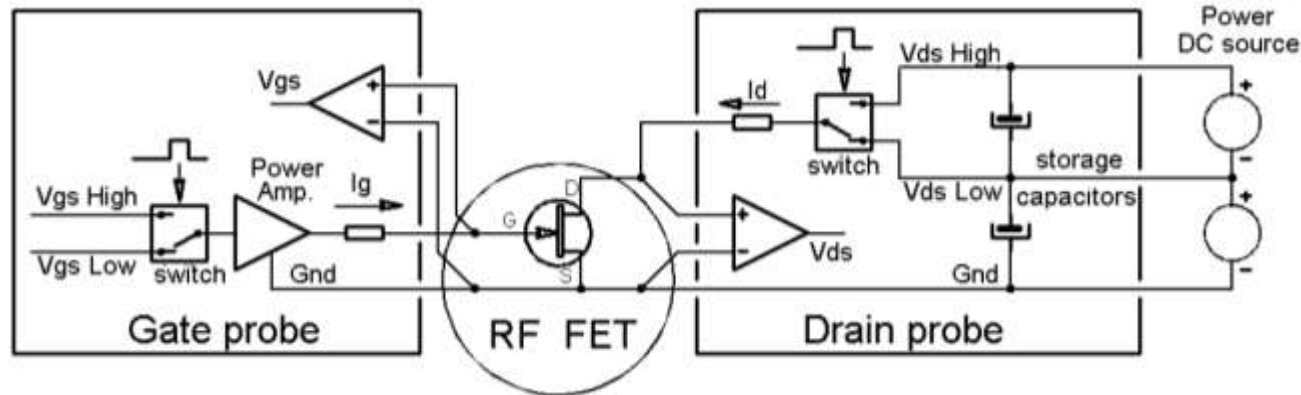
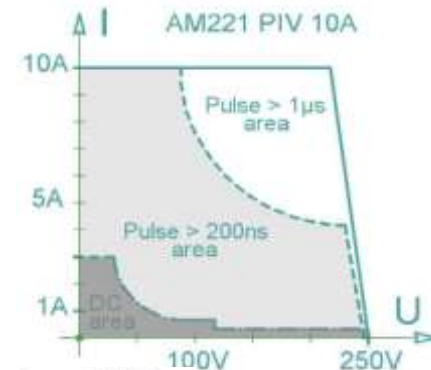
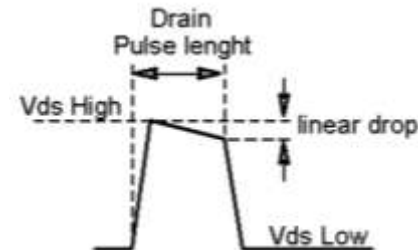
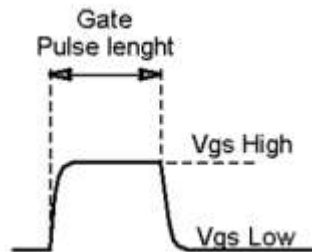
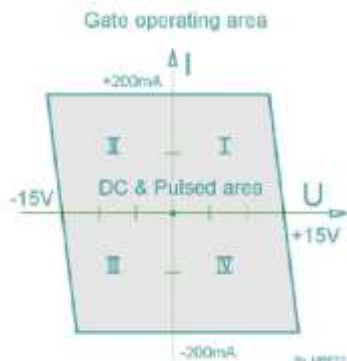




Pulsed IV parameter measurements

How to get accurate pulsed IV measurements ?

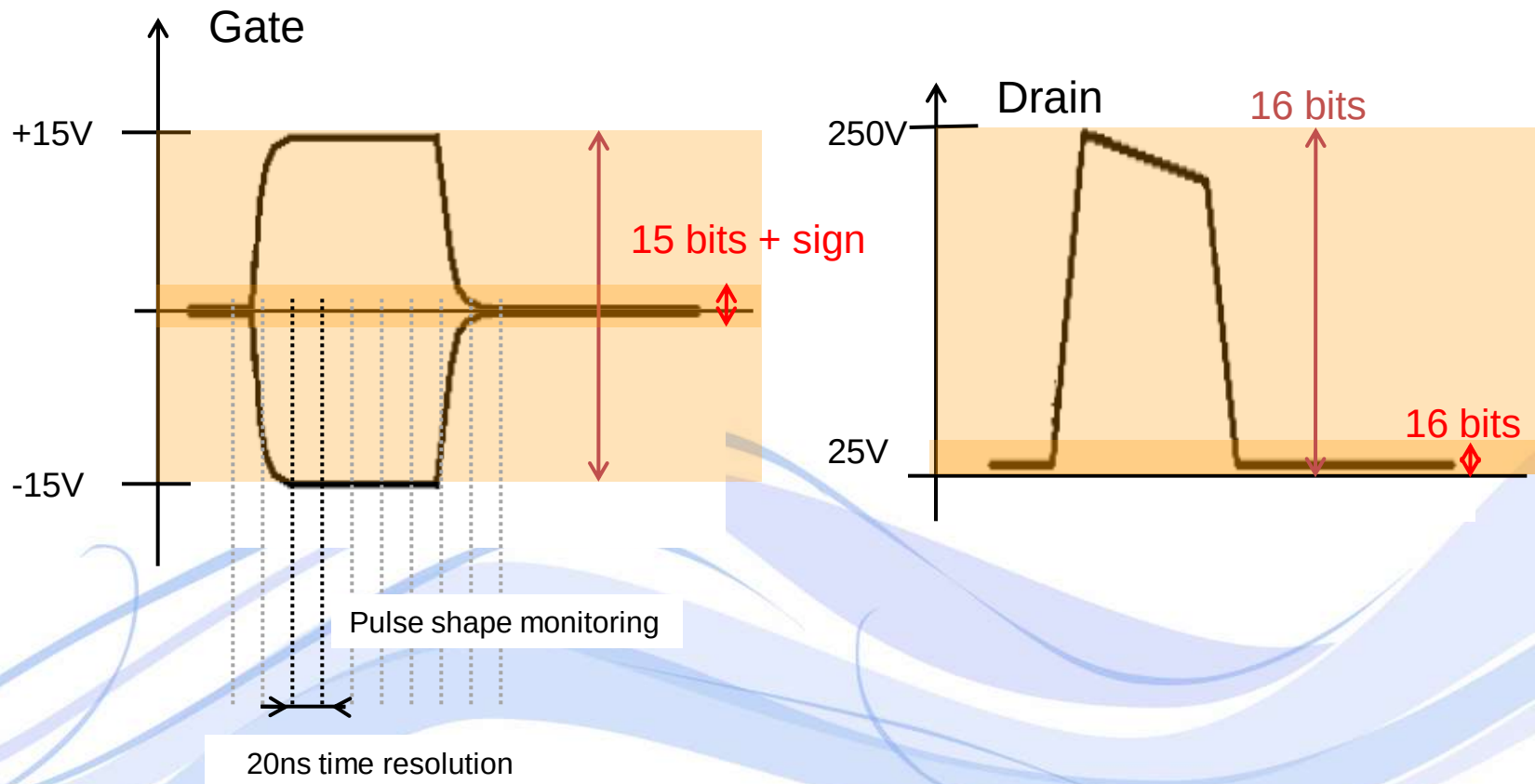
➤ PIV system





Pulsed IV parameter measurements

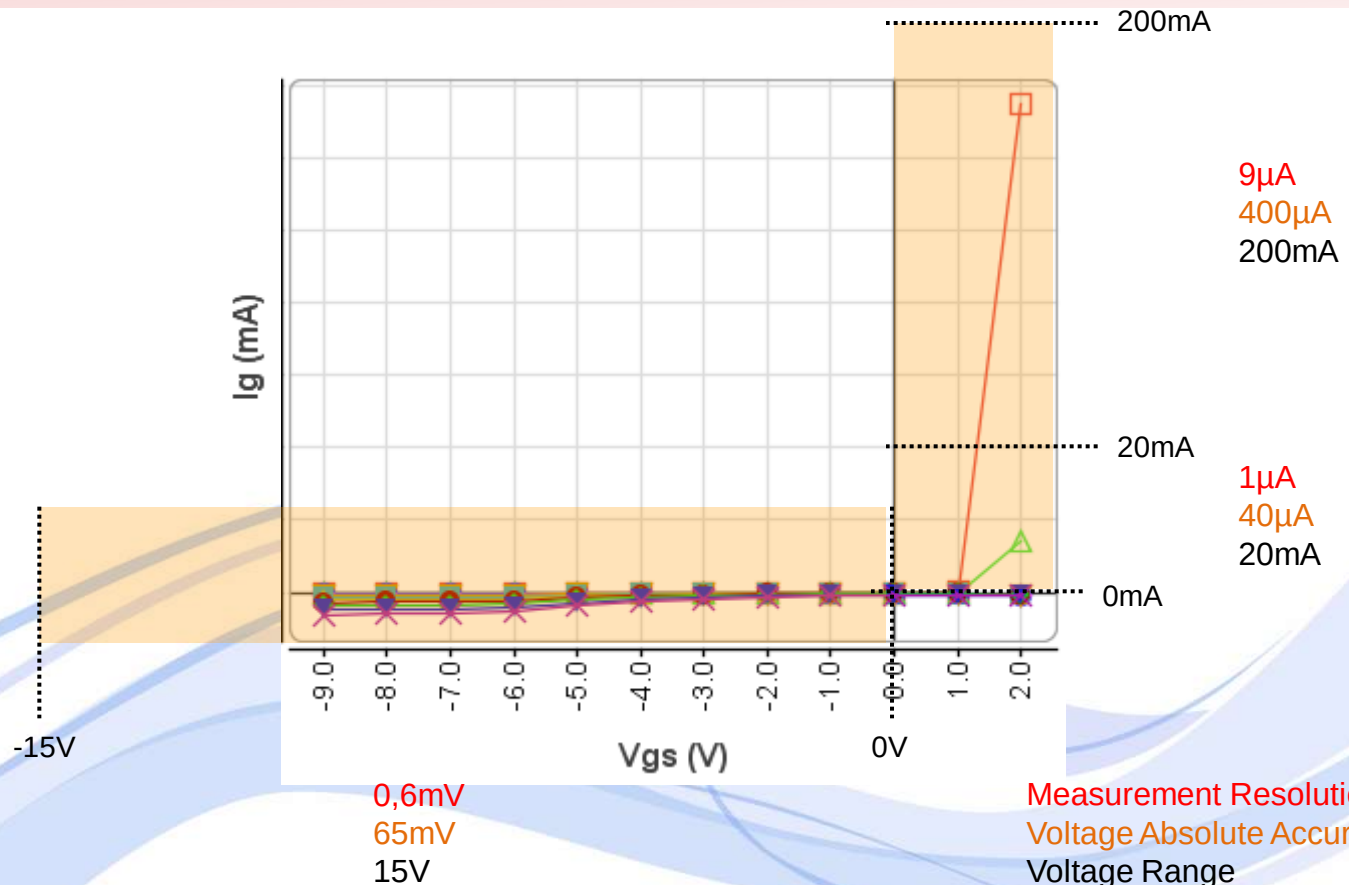
How to get accurate pulsed IV measurements ?





Pulsed IV parameter measurements

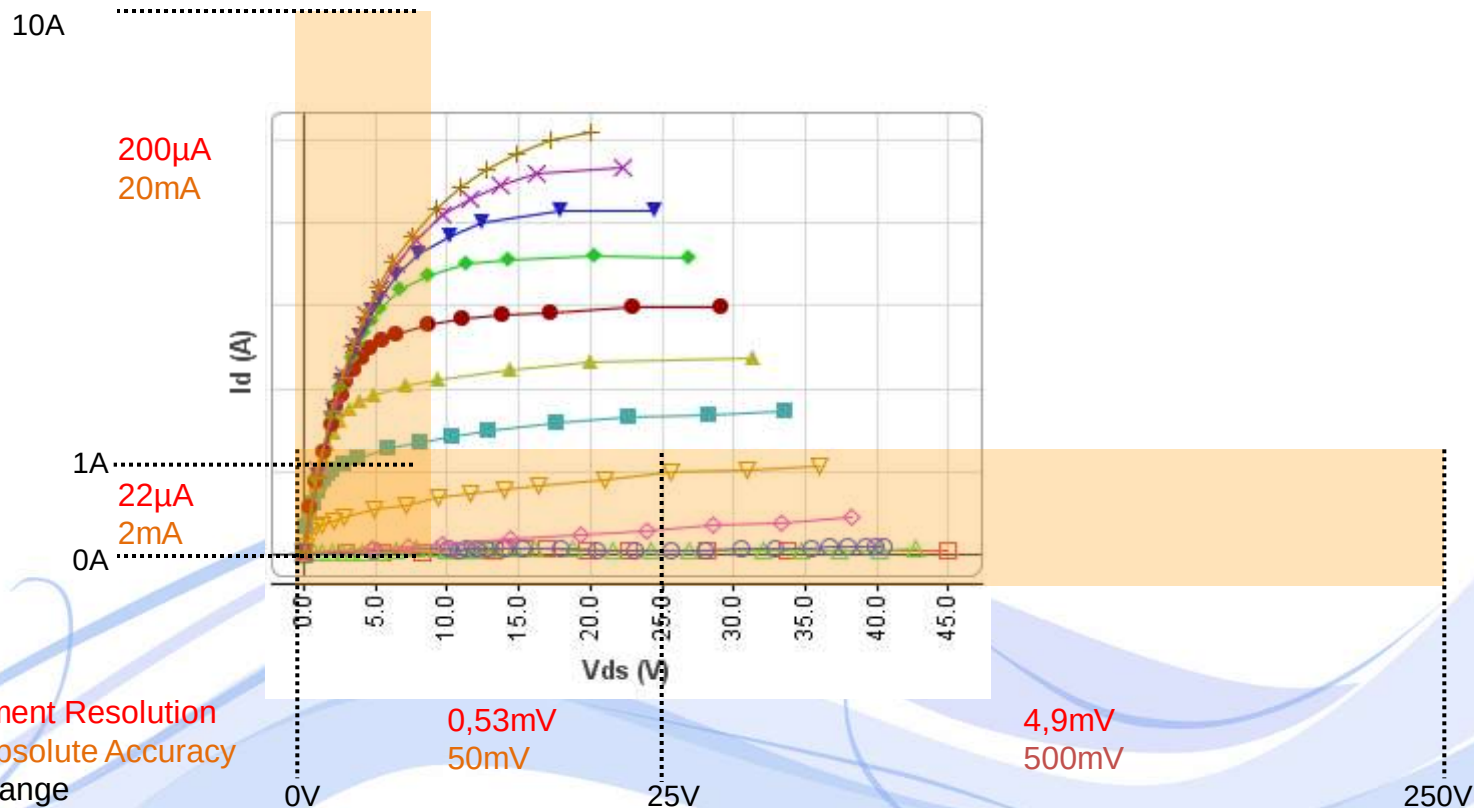
How to get accurate pulsed IV measurements ?





Pulsed IV parameter measurements

How to get accurate pulsed IV measurements ?





Agilent Technologies

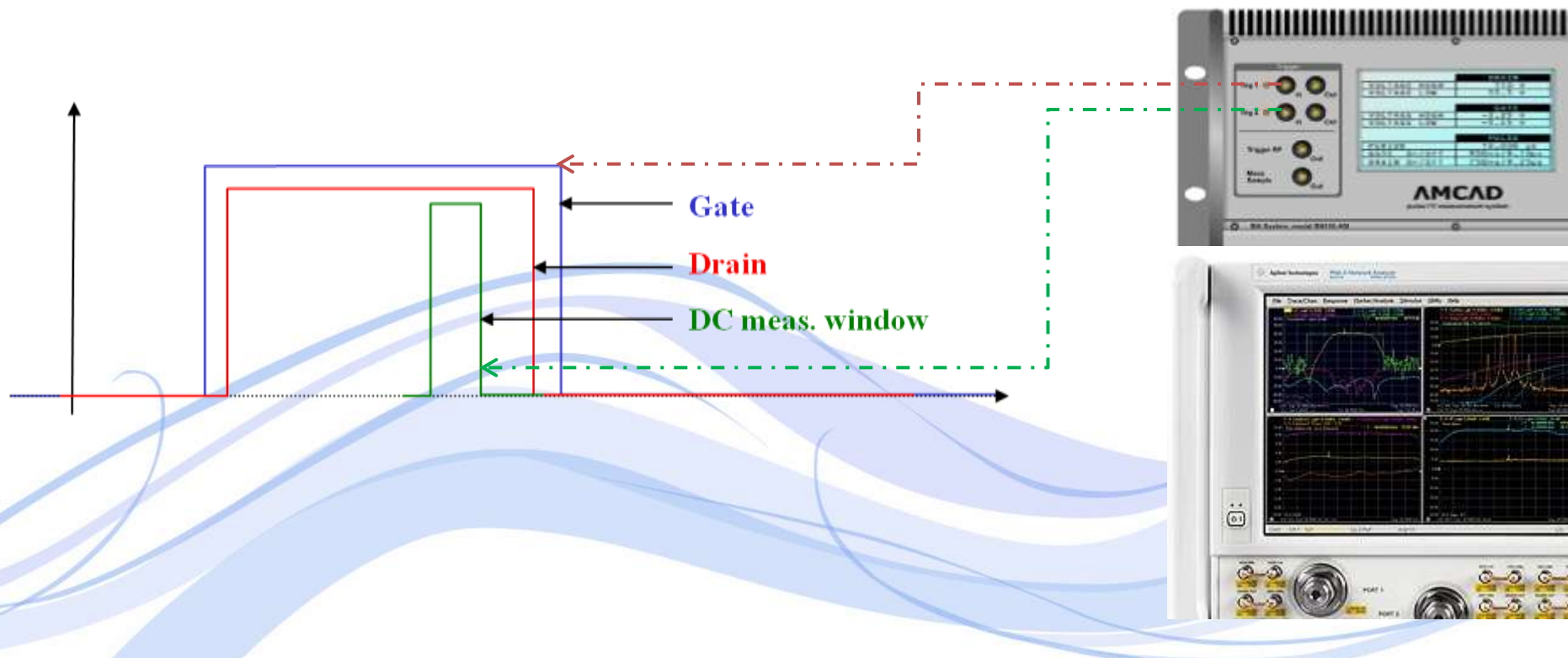


AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Pulsed IV & S parameter measurements

How to get accurate pulsed IV measurements ?

Synchronisation between Pulse IV and pulse S parameters



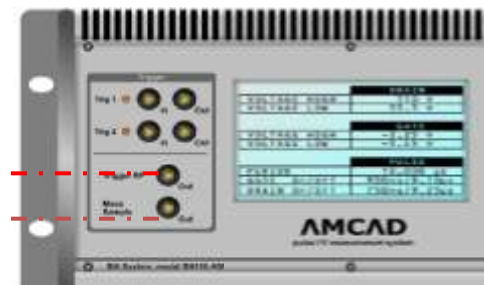
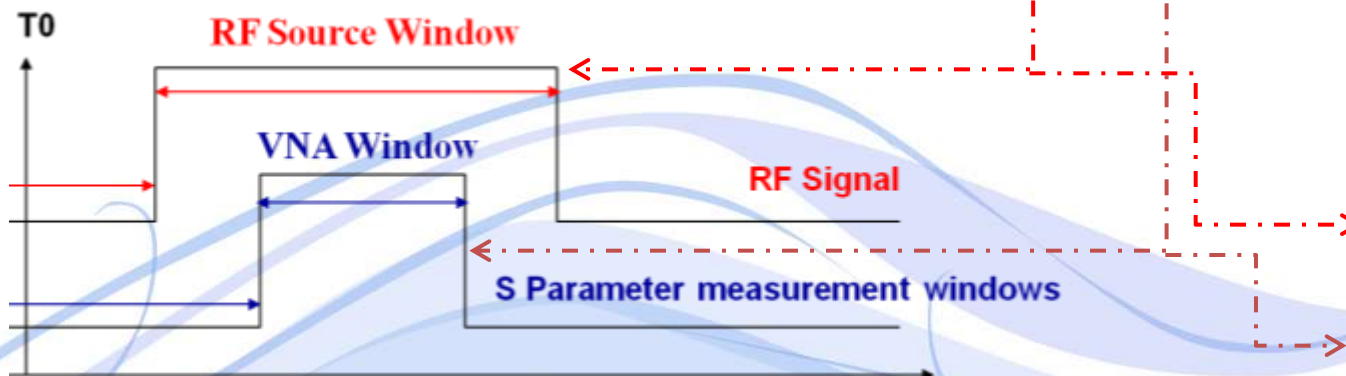
Your complete measurement & modeling solutions partner!



Pulsed S parameter measurements

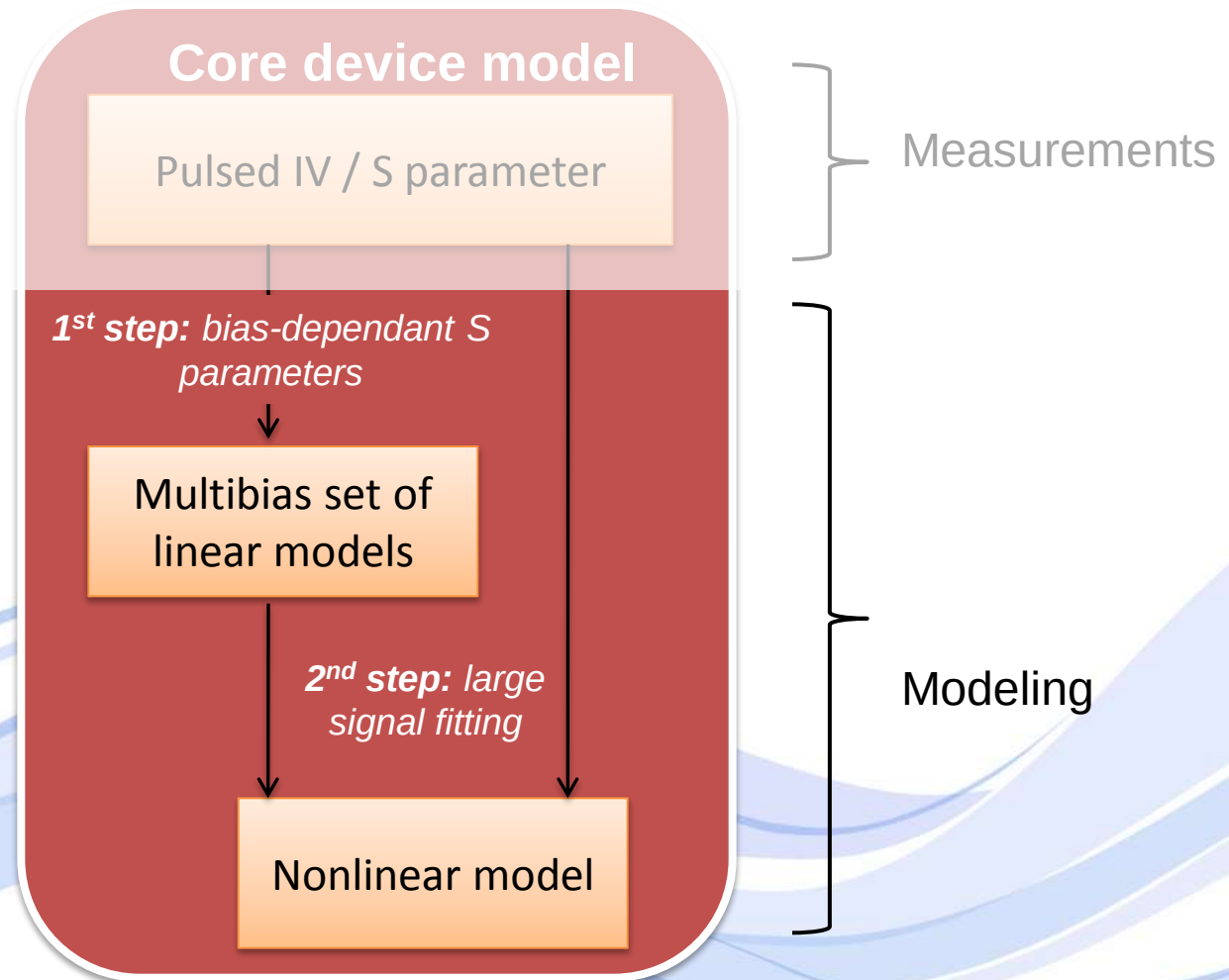
How to get accurate pulsed IV measurements ?

Synchronisation between Pulse IV and pulse S parameters



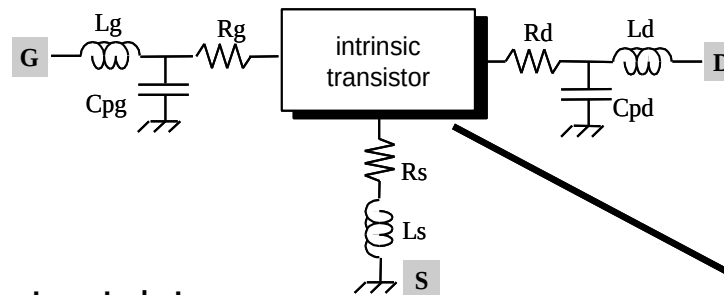


Parameter extraction methodology





Small signal FET modeling

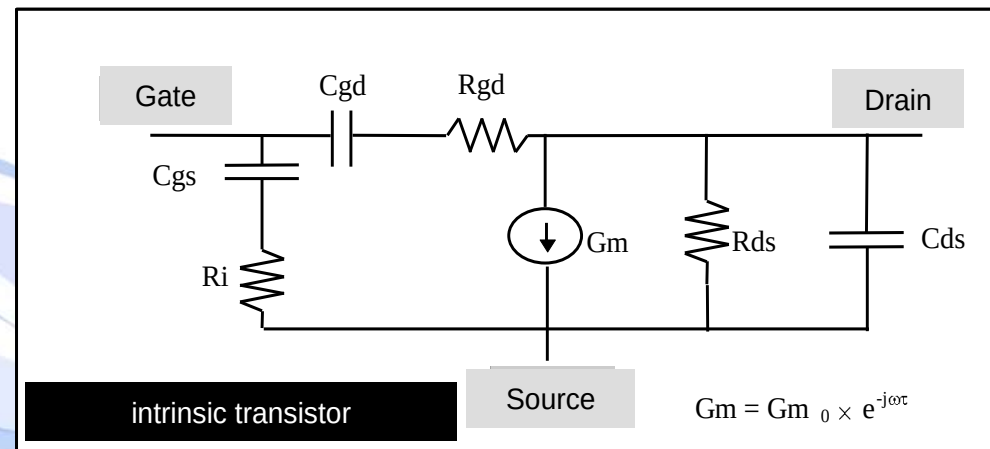


• Extrinsic parameters

- pad capacitances C_{pg} , C_{pd}
- port metallisation inductances L_g , L_d , L_s
- port ohmic resistances R_g , R_d , R_s

• Intrinsic parameters

- channel capacitances C_{gs} , C_{gd}
- voltage-controlled current source with transconductance g_m and transit time delay τ
- ohmic resistances R_i , R_{gd}
- output capacitance C_{ds} and resistance R_{ds}



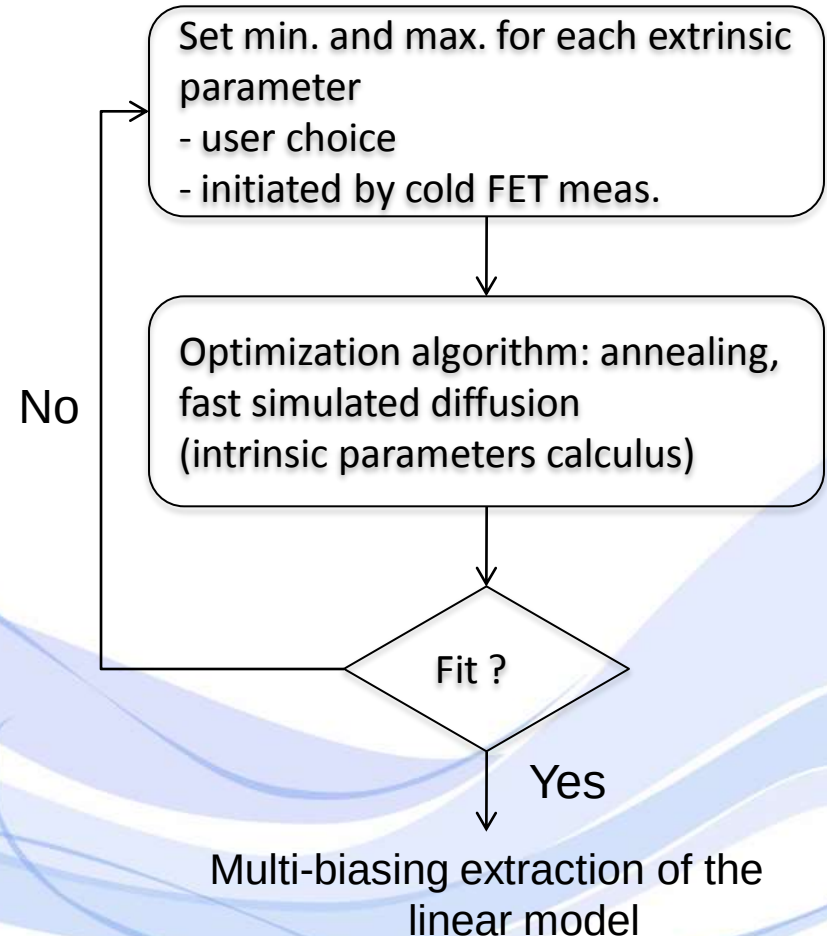


Small signal FET modeling

- Extraction of extrinsic and intrinsic parameters:

There is only one set of extrinsic parameters for which intrinsic parameters are independent from the frequency

For a given set of extrinsic parameters, intrinsic admittance matrix of the device is extracted from measured [S] parameters





Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

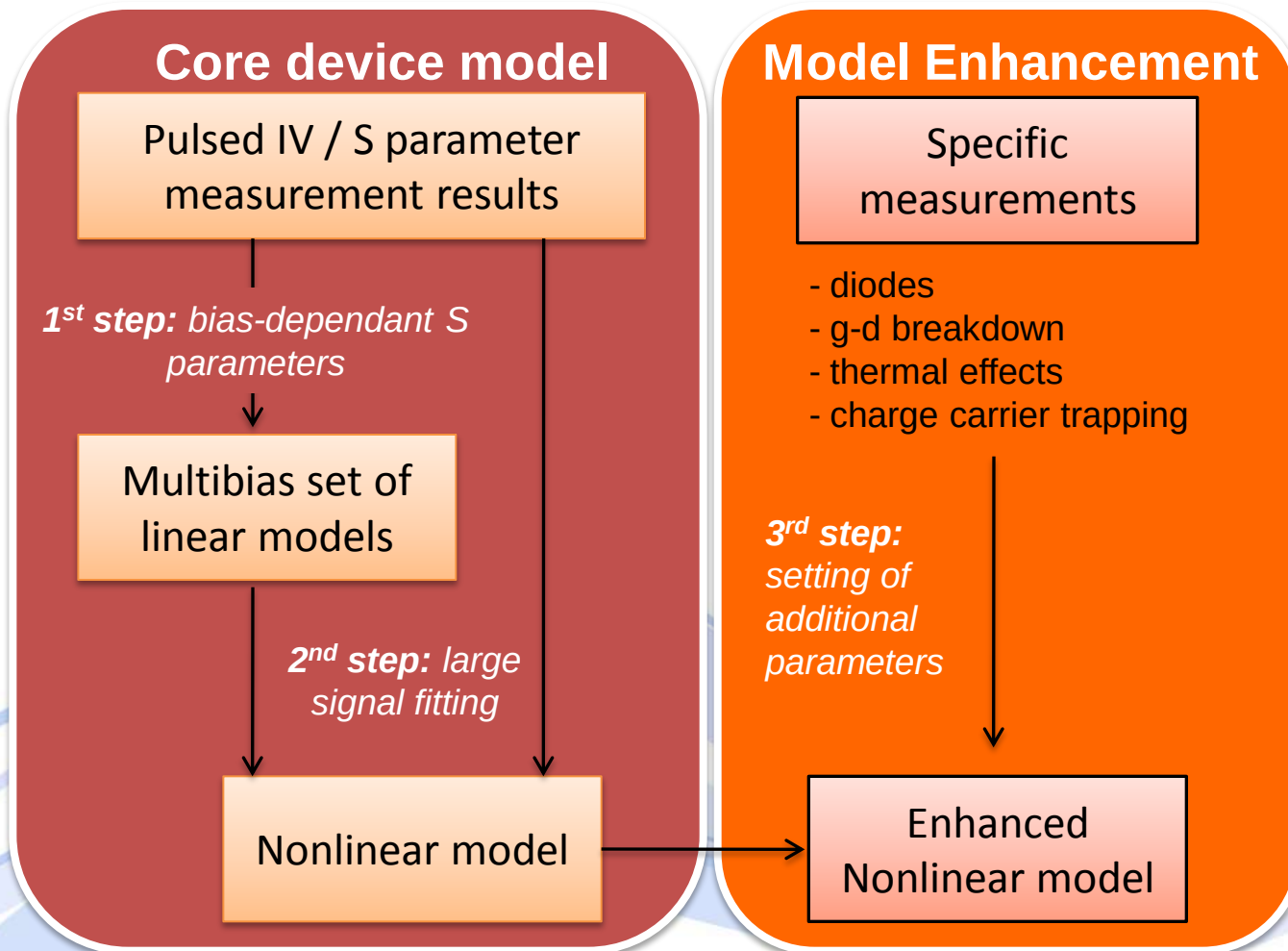
OUTLINE

- Introduction
- Core device model extraction
- Model Enhancement
- Model Validation

Your complete measurement & modeling solutions partner!

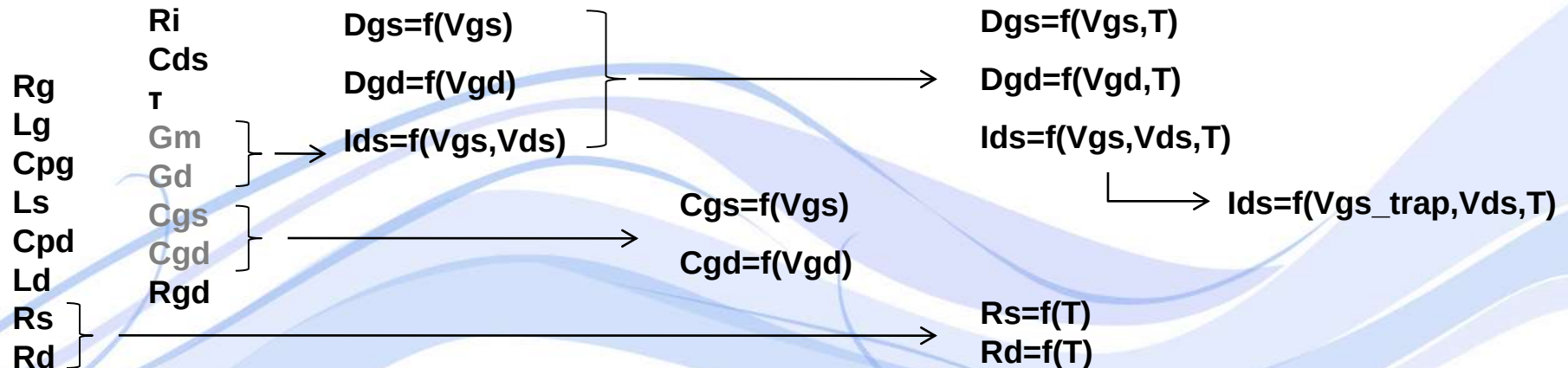
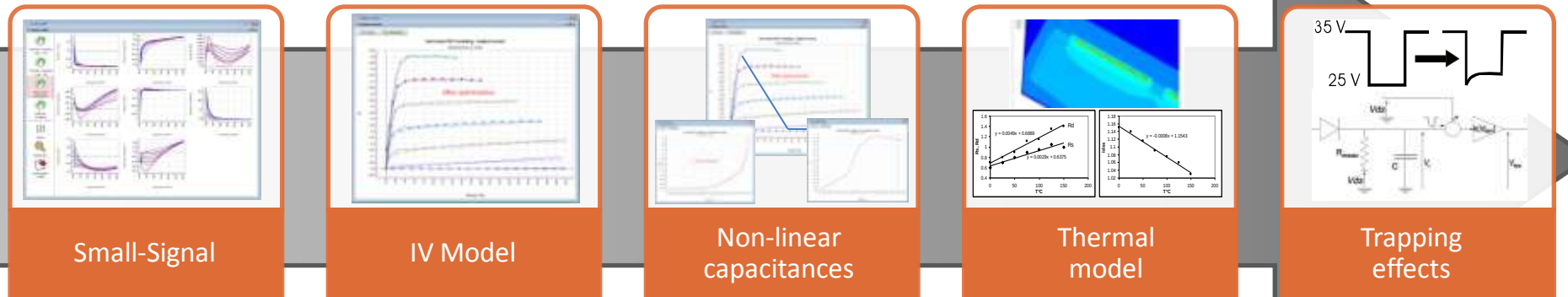


Parameter extraction methodology





Compact FET model extraction flow



Various effects are successively added



Non linear capacitances

- Gate charge is partitioned into gate source and gate-drain charge. Each charge expression is a function of both V_{DS} and V_{GS} .
- For power amplifier applications, 2D models do not bring a breakthrough in the precision, but they are much more complex
- 1 dimension capacitances extracted along optimal load-line are preferred due to simplicity.
1D capacitances with equations based on hyperbolic tangents are naturally charge conservatives
- Output Capacitance C_{ds} is linear – no voltage dependence (weak anyway)
 - $C_{gd}=f(V_{gd})$
 - + Modeling simplicity. Very good convergence
 - Validity of the 1D ?
 - $C_{gs}=f(V_{gs})$

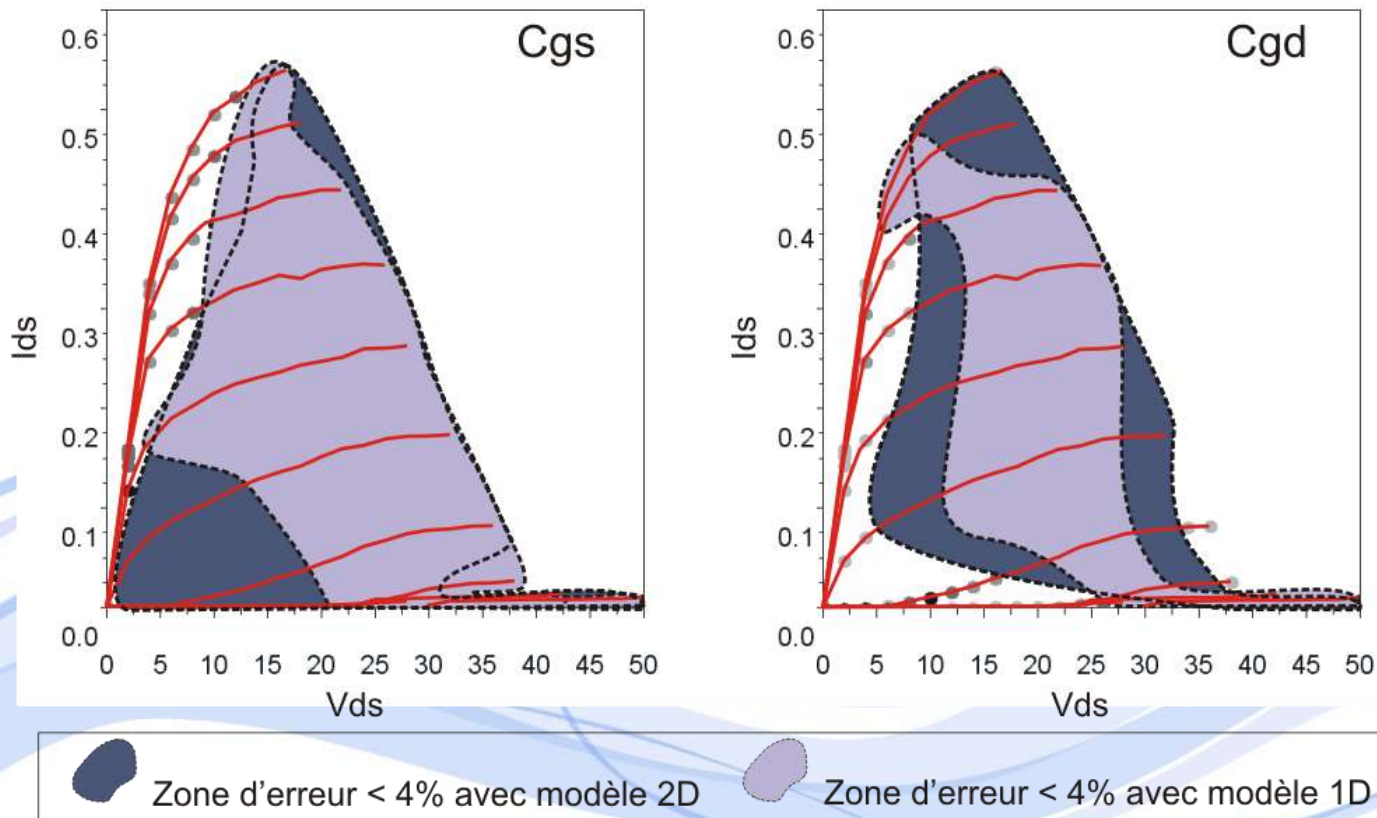


Non linear capacitances

- Capacitances modeling: 1D vs 2D

→ Better fit of the 2D model in wider domain

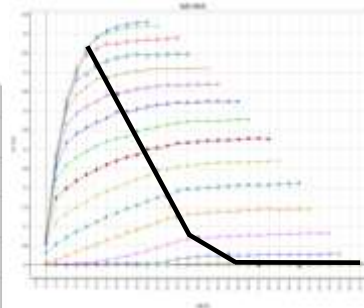
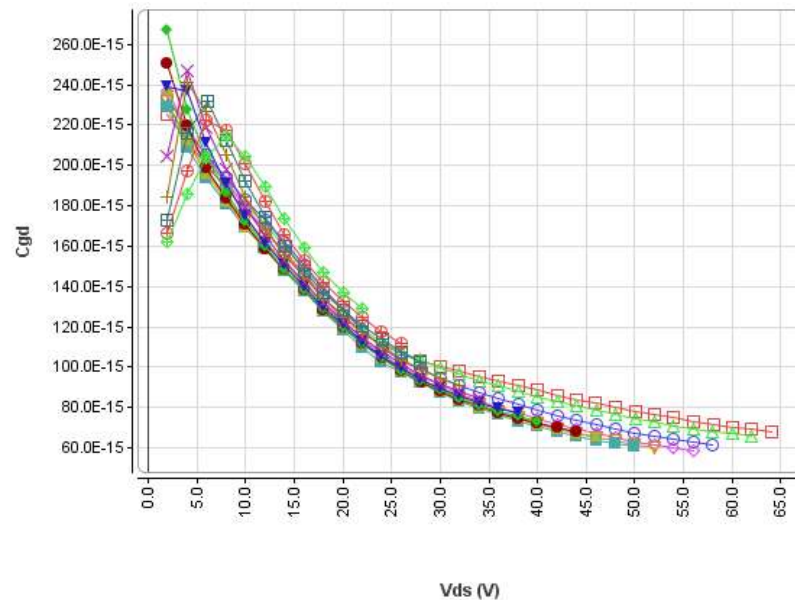
→ But the 1D model has a good behavior



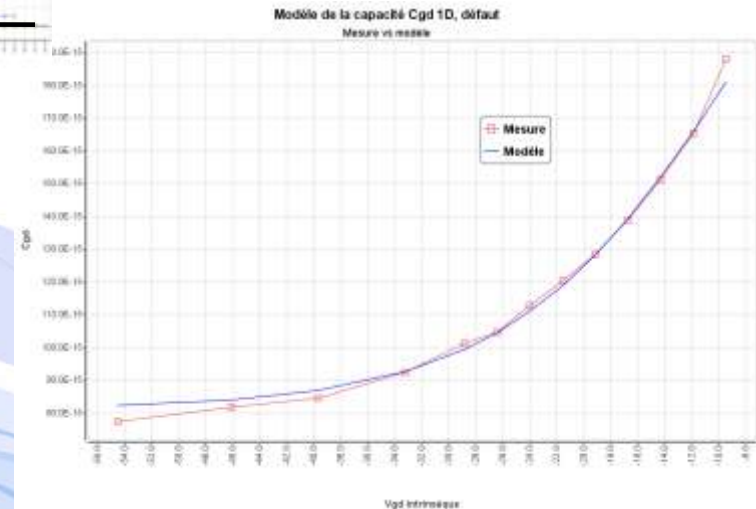


Non linear capacitances

- Cgd
- Feedback capacitance Cgd is a strong function of drain voltage.



Cgd capacitance extracted along optimal load-line for power amplification



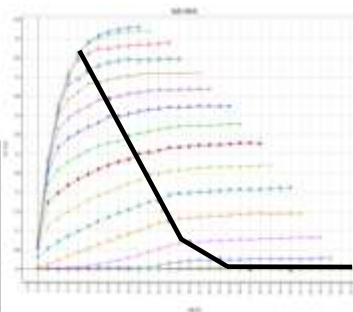
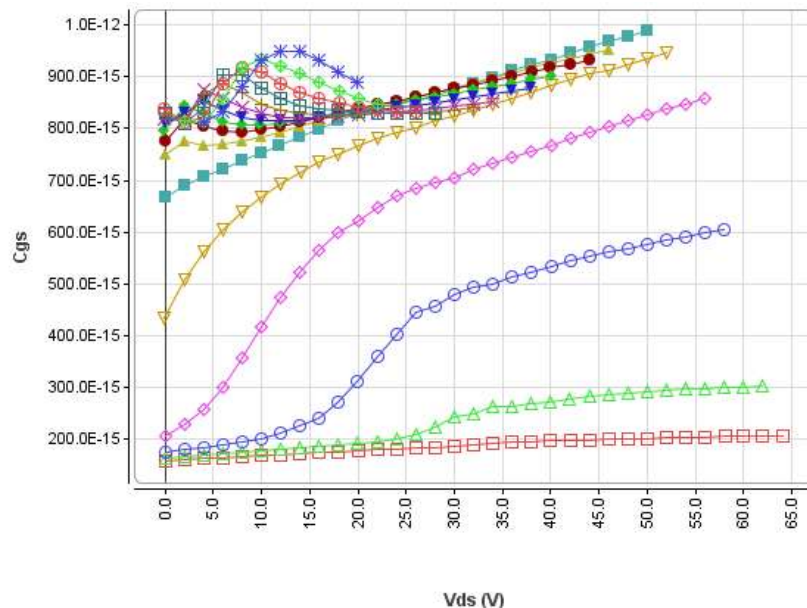


Non linear capacitances

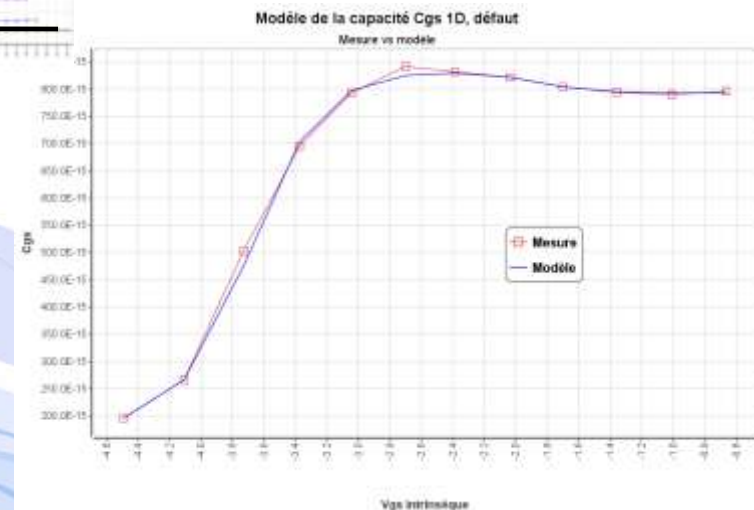
C_{gs}

- Input capacitance C_{gs} is a strong function of gate voltage.

The gate-voltage non-linearity also effects model's harmonic generation

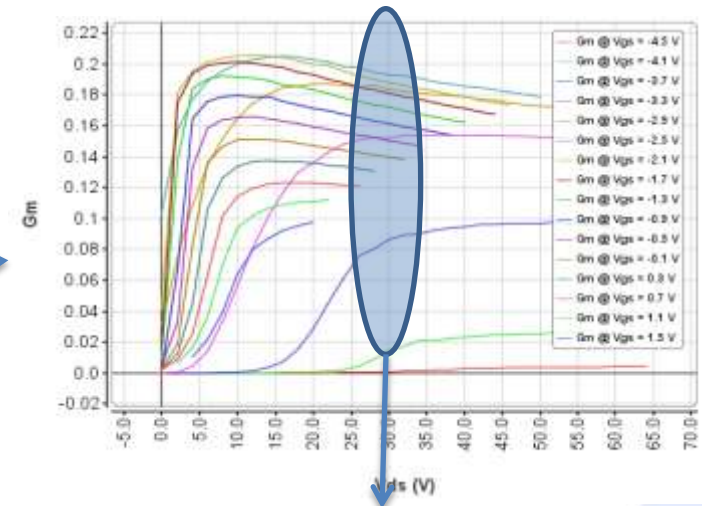
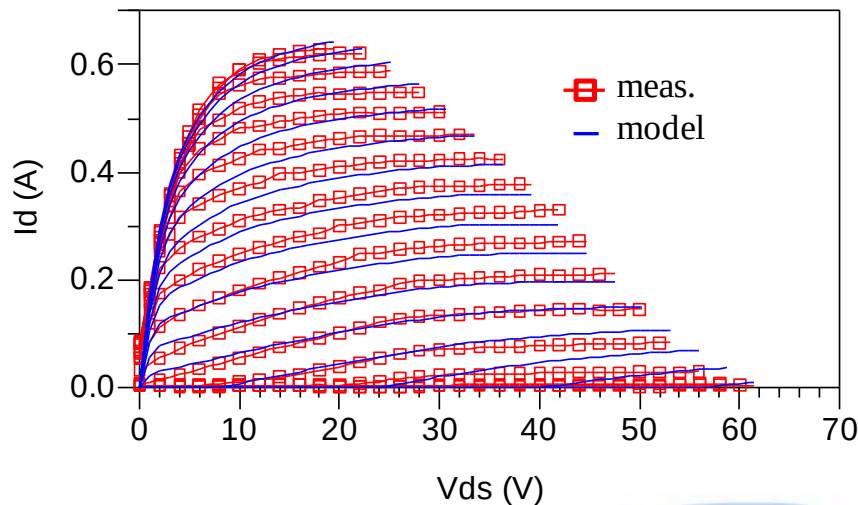


C_{gs} capacitance extracted along optimal load-line for power amplification

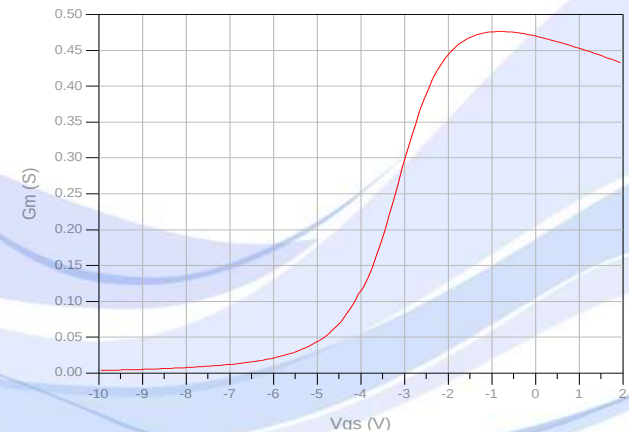




Output current source

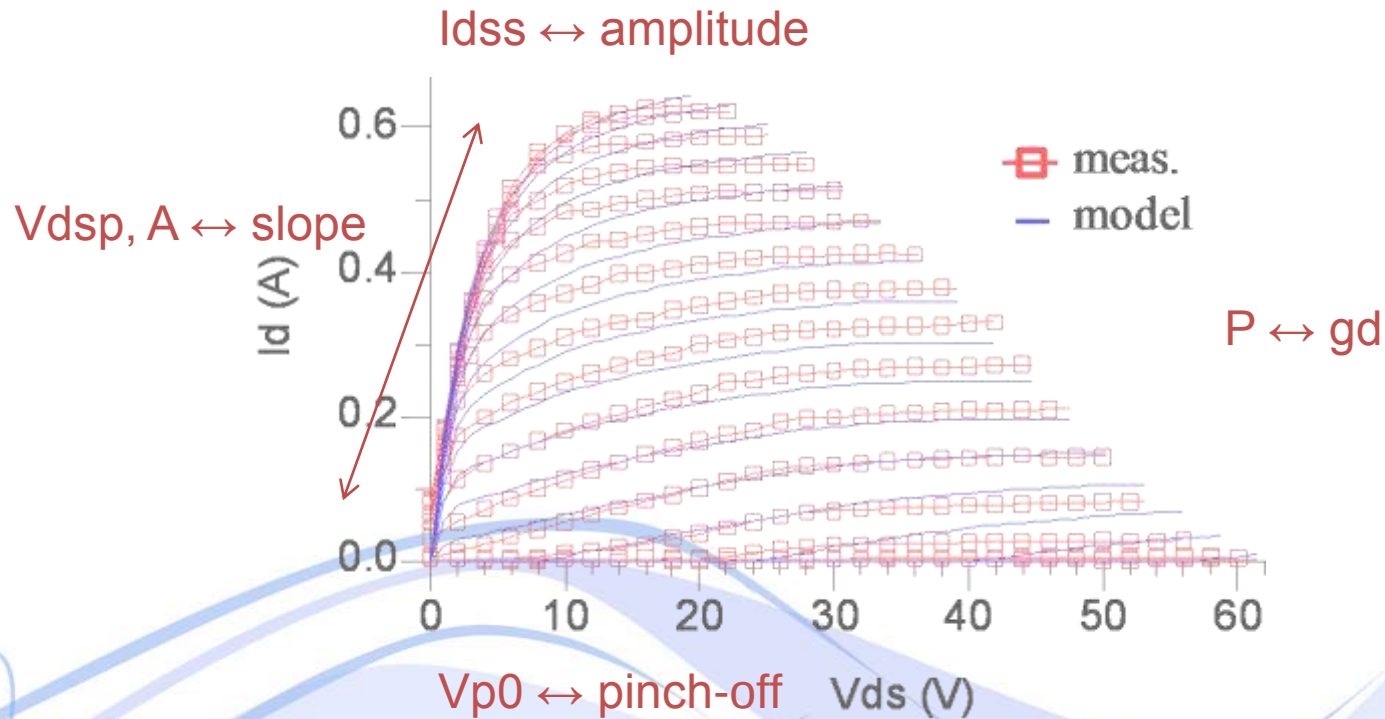


- AMCAD drain current model formulation allows to predict very accurately the I-V curves, the partial derivatives g_m and g_d , the knee voltage and the transconductance decrease at high current.





Output current source



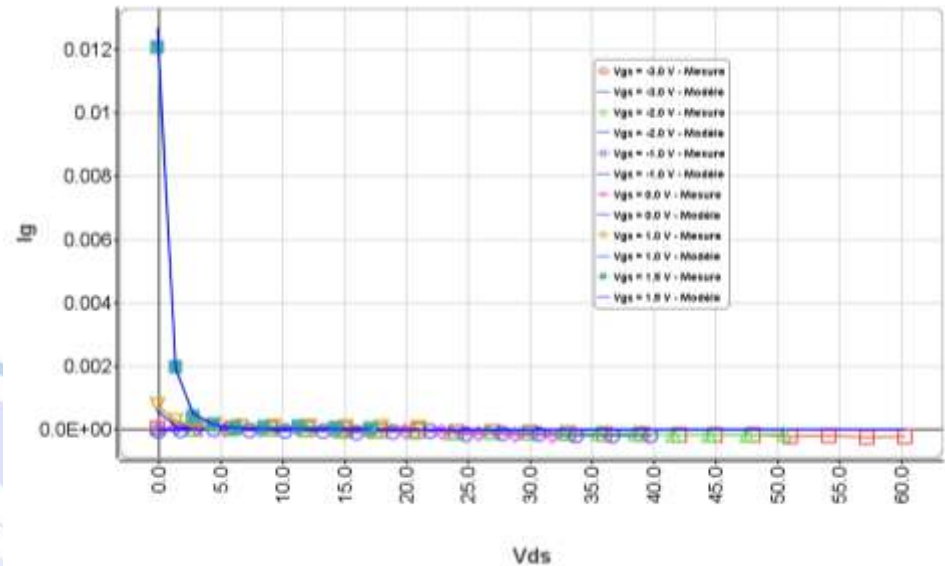
$M, P \leftrightarrow$ fitting parameters

$\text{AlphaGm}, V_{gm}, \text{BetaGm}, V_{dm} \leftrightarrow g_m$ (derivative)

Diodes

- Gate-drain and gate-source diode equations include forward conduction of gate current

$$I_D = I_s e^{\text{Alpha} \cdot V_d} \text{ (a classical formulation)}$$

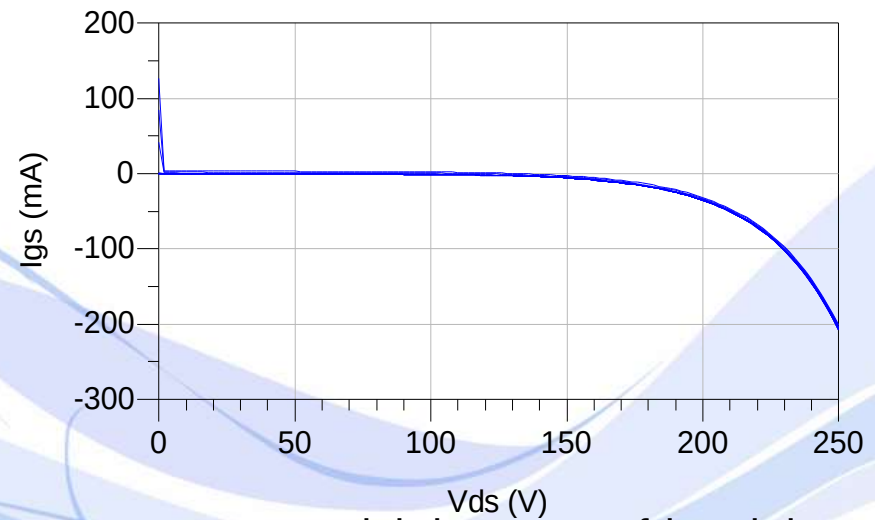
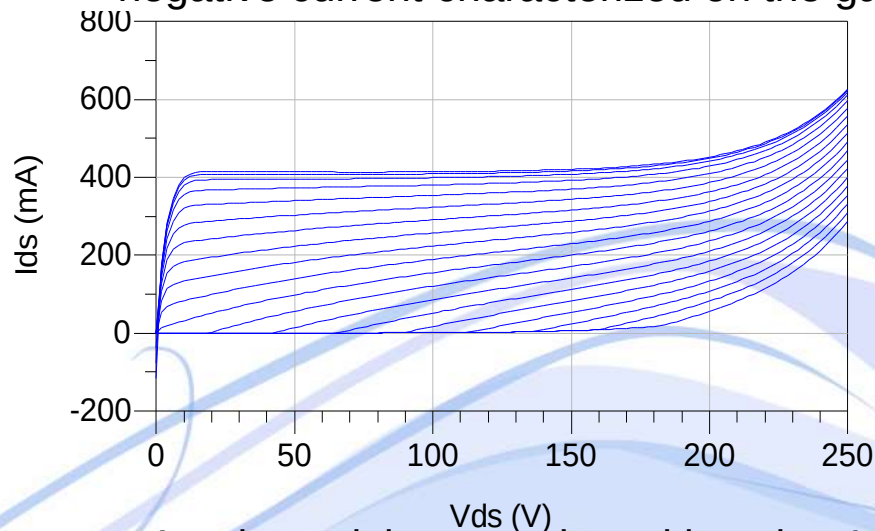




Breakdown generator

- Gate-drain Breakdown generator

The breakdown phenomena leads to a current from the drain to the gate when the device is pinched-off and for high values of V_{ds} voltage. In this case, the whole negative current characterized on the gate is seen in positive on the drain

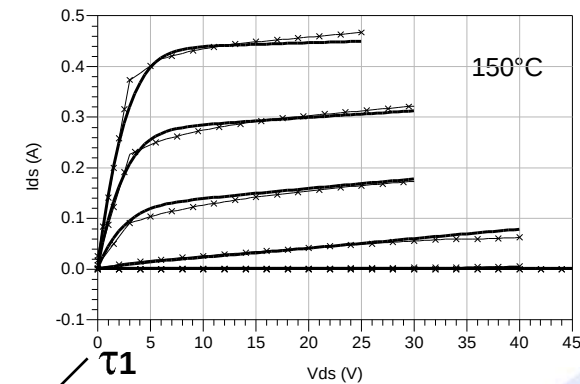
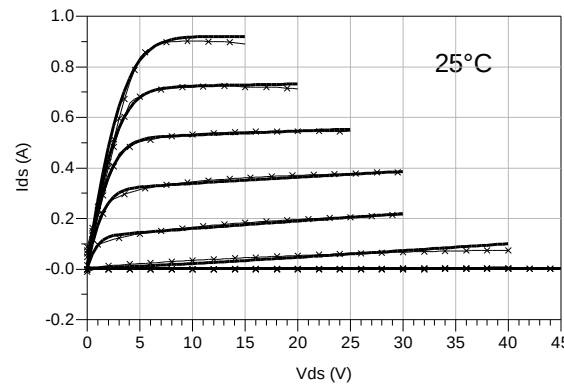
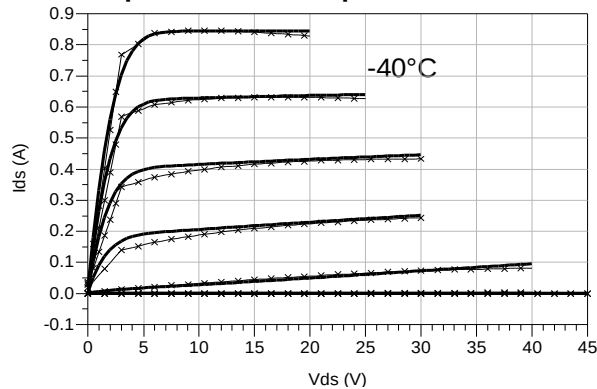


A polynomial expression with order 4 is necessary to model the cross of breakdown curves, with varies depending on the process

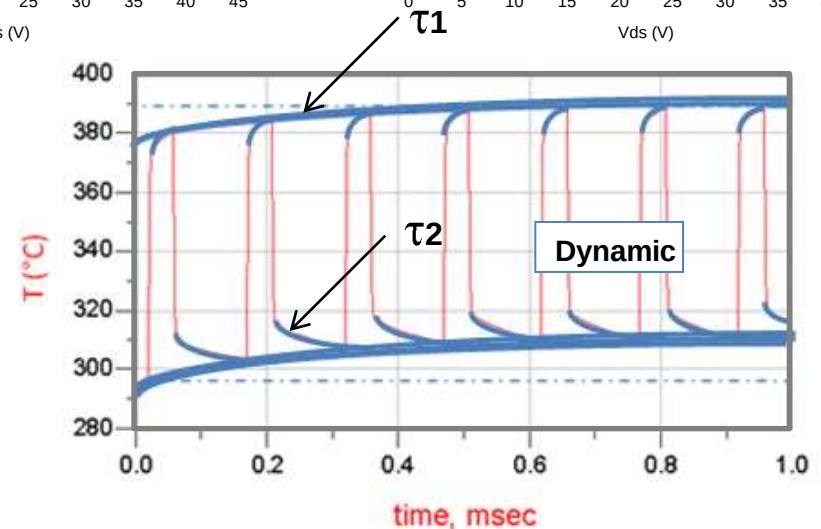
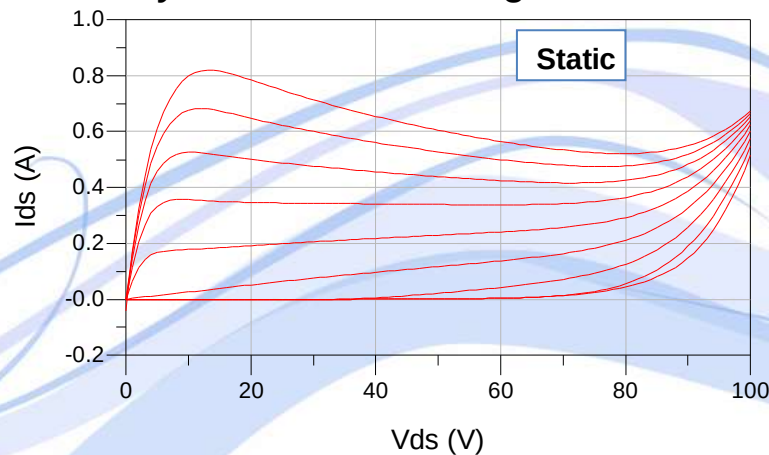


Thermal effects

- Temperature dependence with ambient or chuck temperature



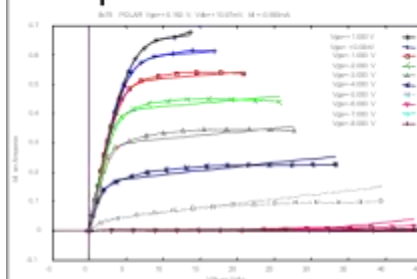
- Static and Dynamic self-heating effects



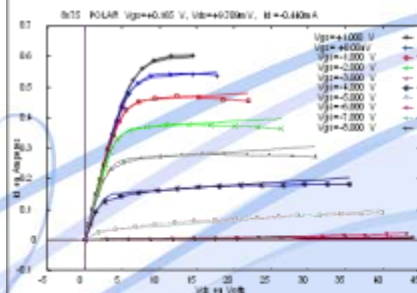


Thermal effects

IV Modeling @ several temperatures

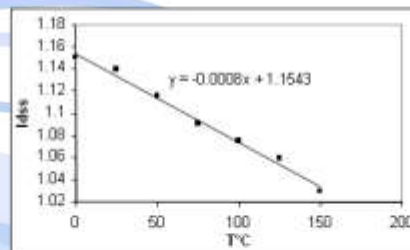
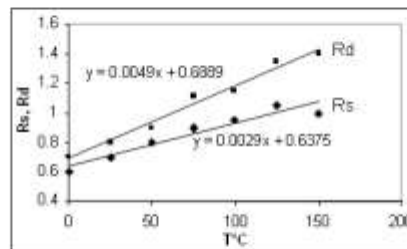


Measure / IV Model @ 25 °C



Measure / IV Model @ 150 °C

IV parameters versus temperature



Temperature dependence

Equations

Thermal parameters

- Access Resistances
- Current Source
- Diodes

$$R_s = R_{s0} + \alpha_{R_s} \cdot T$$

$$R_d = R_{d0} + \alpha_{R_d} \cdot T$$

$$I_{dss} = I_{dss0} + I_{dss_t} \cdot T$$

$$P = P_0 + P_t \cdot T$$

$$N_{gs} = N_{gs0} + N_{gs_t} \cdot T$$

$$N_{gd} = N_{gd0} + N_{gd_t} \cdot T$$

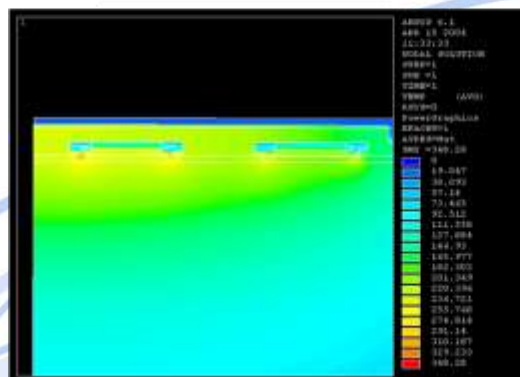
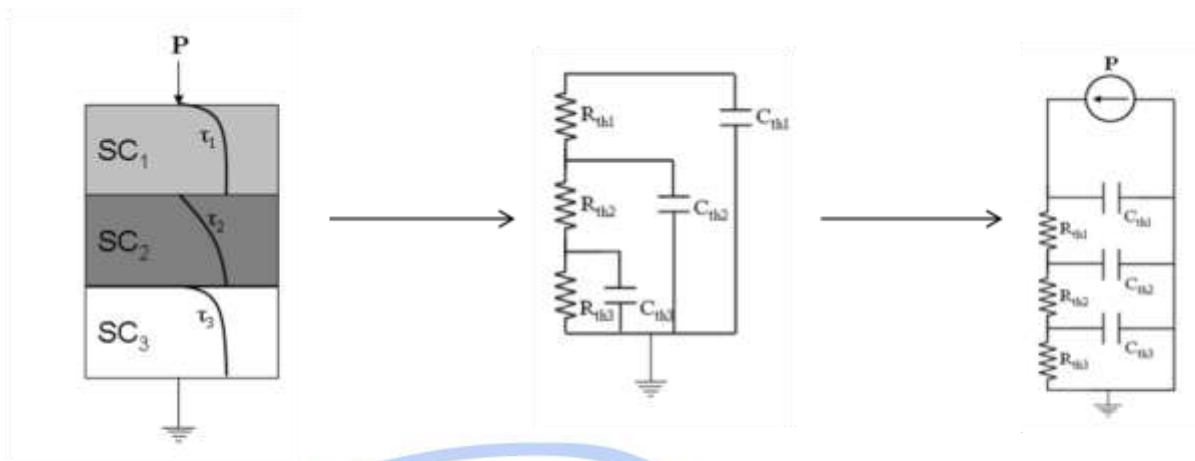
$$I_{sgs} = I_{sgs0} + I_{sgs_t} \cdot e^{(T/T_{sgs})}$$

$$I_{sgd} = I_{sgd0} + I_{sgd_t} \cdot e^{(T/T_{sgd})}$$



Thermal effects

Thermal impedance extraction – *by simulation*



Thermal simulations

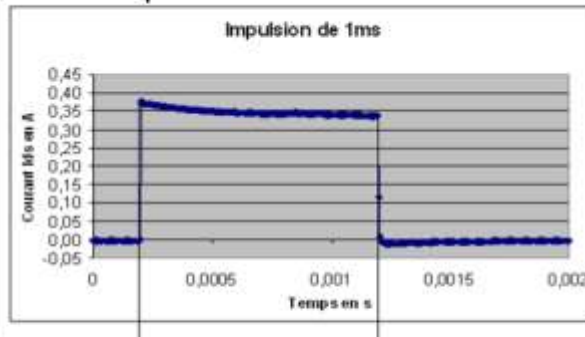
$$Z_{th}(s) = \frac{1}{sC_{th1} + \frac{1}{R_{th1} + \frac{1}{sC_{th2} + \frac{1}{R_{th2} + \frac{1}{sC_{th3} + \dots + \frac{1}{R_{thn}}}}}}}$$



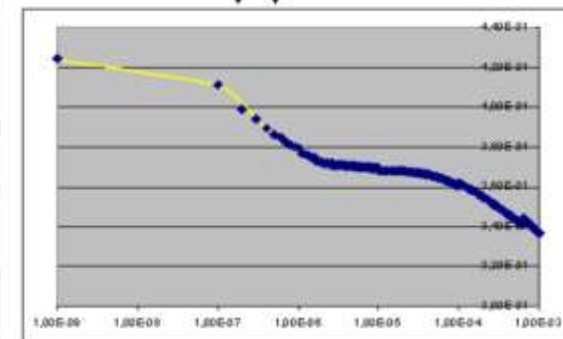
Thermal effects

Thermal impedance extraction – *by measurements*

Wide current pulse characterization



in log scale to determine time constants

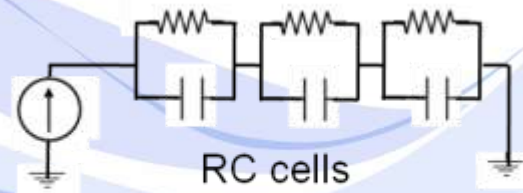


Self Heating Extraction = f(t)

$$i(t) = I_0 - \sum_{i=1}^n I_i \cdot \left(1 - \exp\left(\frac{-t}{\tau_i}\right) \right)$$



$$Z_{th}(t) = \sum_{K=1}^n R_{kth} \left(1 - e^{-\frac{t}{R_{kth} C_{kth}}} \right)$$



Device dynamical self-heating

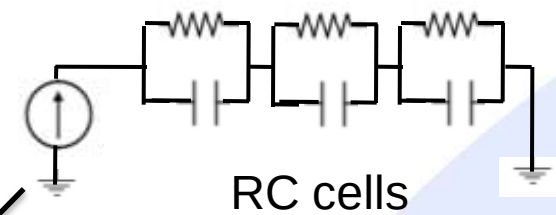
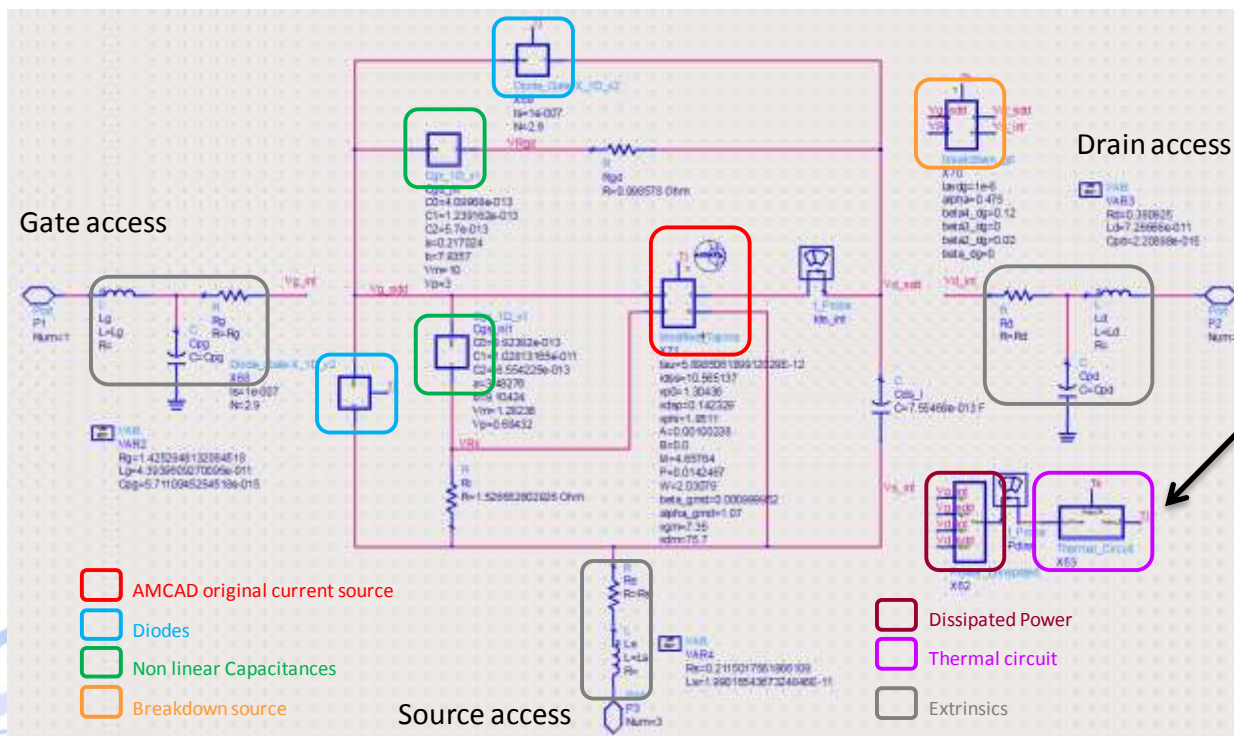


Thermal effects

- Drain current is only temperature dependent model element

Takes into account ambient temperature and self-heating effects

Thermal analog circuit to model self-heating and elevated heat sink temperatures



RC cells

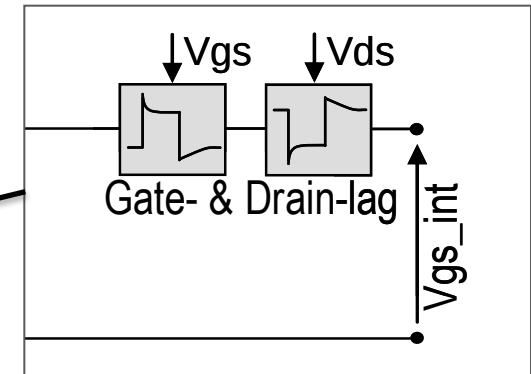
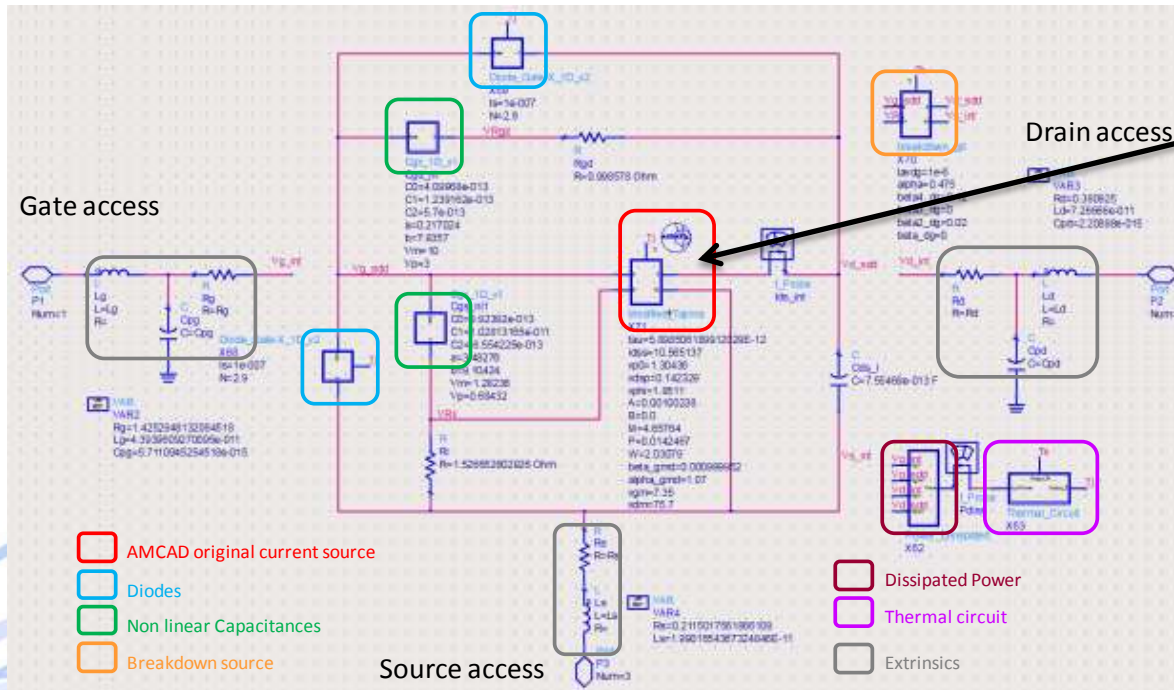
Source access

Drain access



Trapping effects

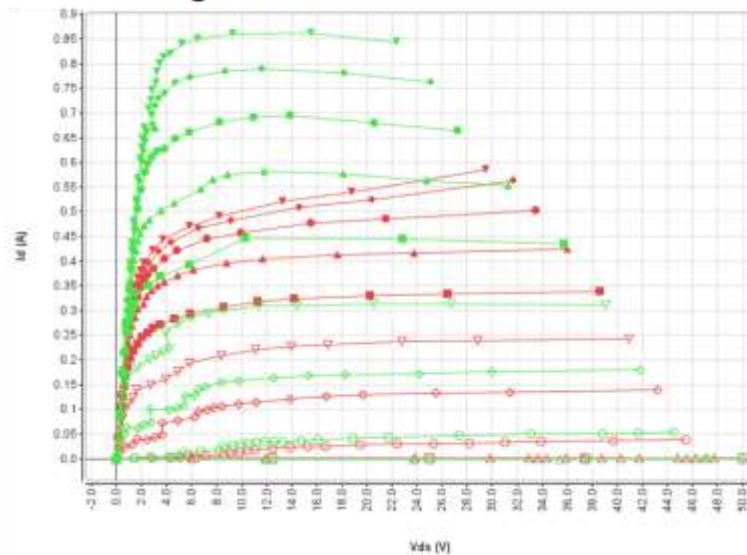
- Charging and discharging of traps has influence on I_{ds} and leads to current collapse. This is described in the model by trapping effects modifying the gate command and separated into gate and drain lag sub-circuits





Trapping effects

Gate-lag: decrease of drain current

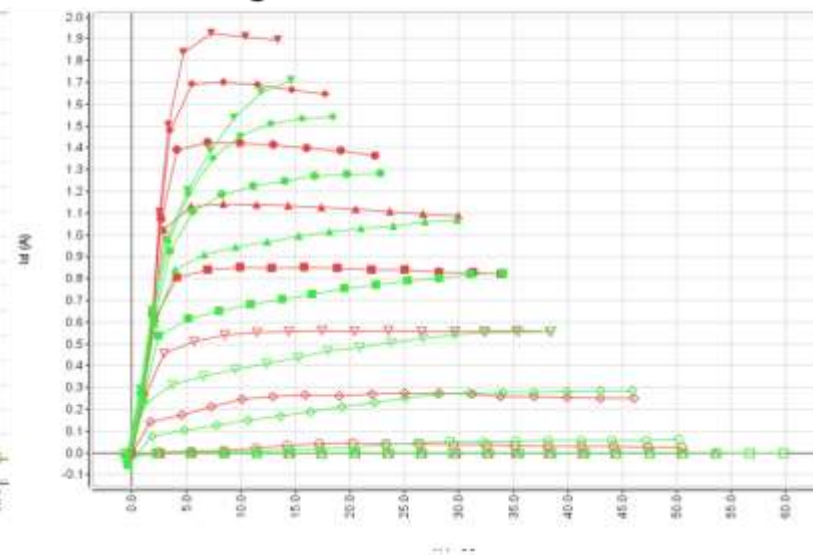


Gate-lag

green ($V_{gs0}=0V, V_{ds0}=0V$)

red ($V_{gs0}=-4V, V_{ds0}=0V$)

Drain-lag: increase of V_{knee}



Drain-lag

red ($V_{gs0}=-3V, V_{ds0}=0V$)

green ($V_{gs0}=-3V, V_{ds0}=30V$)

$$T_{capture} \ll t_{IMPULSION} \ll T_{émission}$$

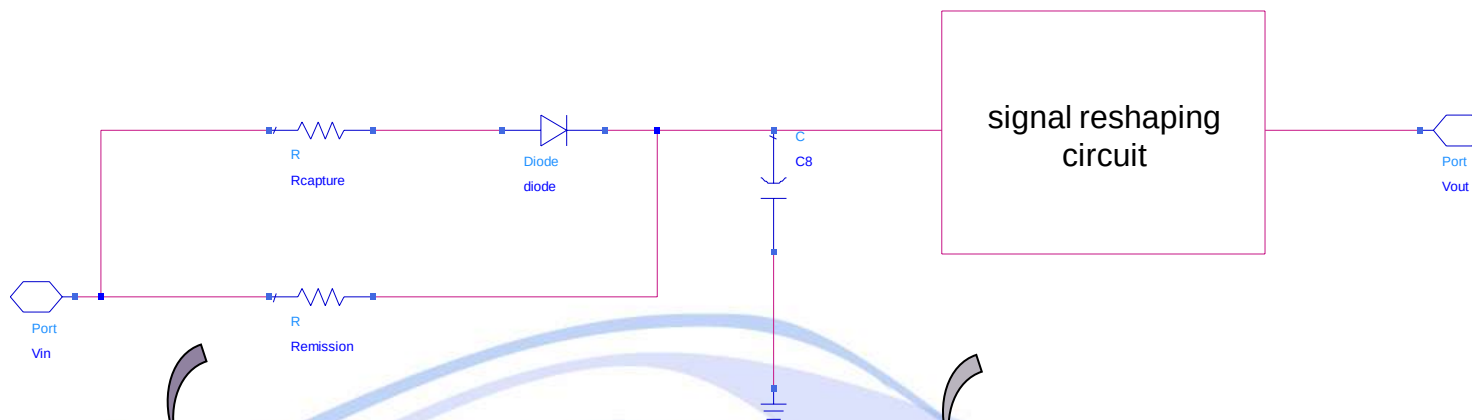
During the pulses capture takes place, emission freezed



Trapping effects

- Charge of the capacitance = Ionized traps

Charge through R_{capture} , Emission through R_{emission}



Diode = dissymmetry of the capture and emission process

Tuning of the magnitude of the trapping effects

Fundamental assumption : dissymmetry of the capture and emission process

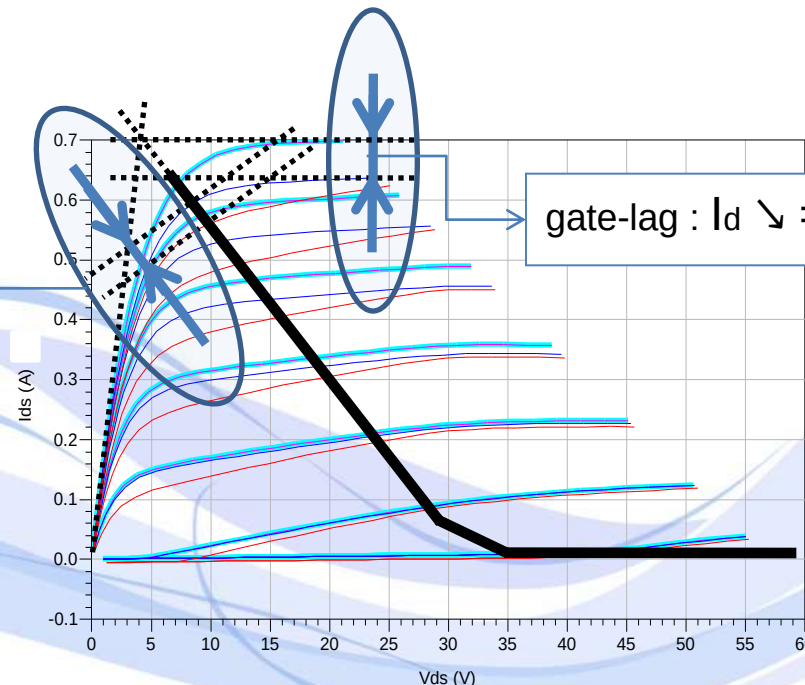


Trapping effects

- Bias dependant gate lag -> current reduction over the entire characteristic

Bias dependent drain lag -> current reduction and shifts the knee-voltage to a higher V_{ds}

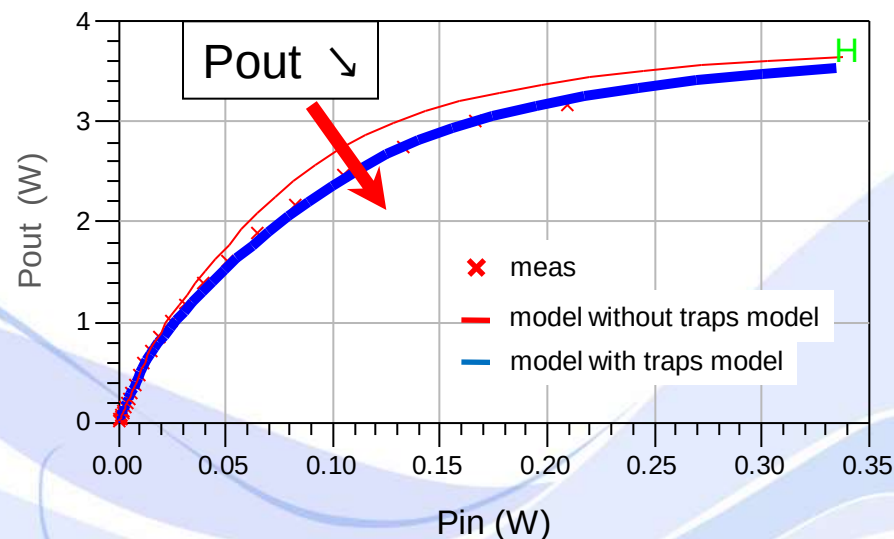
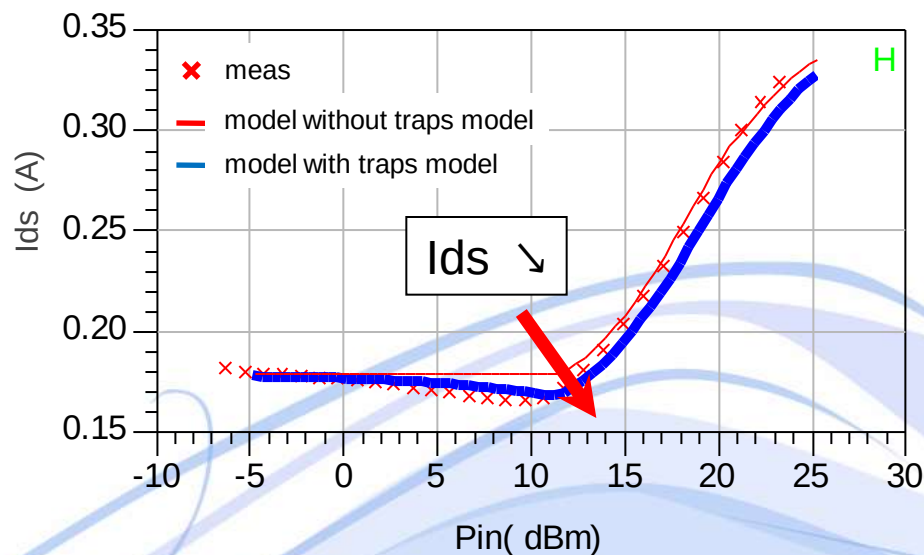
Model covers knee walkout to avoid errors in calculation of output power.





Trapping effects

- Decreasing form of the mean output current only reproduced with traps modeled





Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

OUTLINE

- Introduction
- Core device model extraction
- Model Enhancement
- Model Validation

Your complete measurement & modeling solutions partner!



Parameter extraction methodology

Core device model

Pulsed IV / S parameter
measurement results

*1st step: bias-dependant S
parameters*

Multibias set of
linear models

*2nd step: large
signal fitting*

Nonlinear model

Model Enhancement

Specific
measurements

- diodes
- g-d breakdown
- thermal effects
- charge carrier trapping

*3rd step:
setting of
additional
parameters*

*4th step:
implementation in
commercial simulator*

Enhanced
Nonlinear model

Model Validation

Power
measurements

-load-pull measurements
CW, pulsed
2-tones
time domain

*5th step:
validation and
refinement*

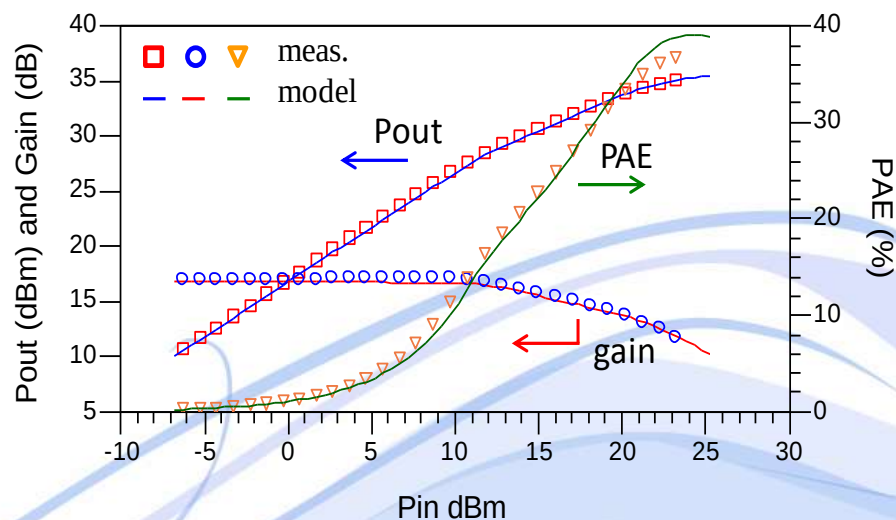
Final Nonlinear
model



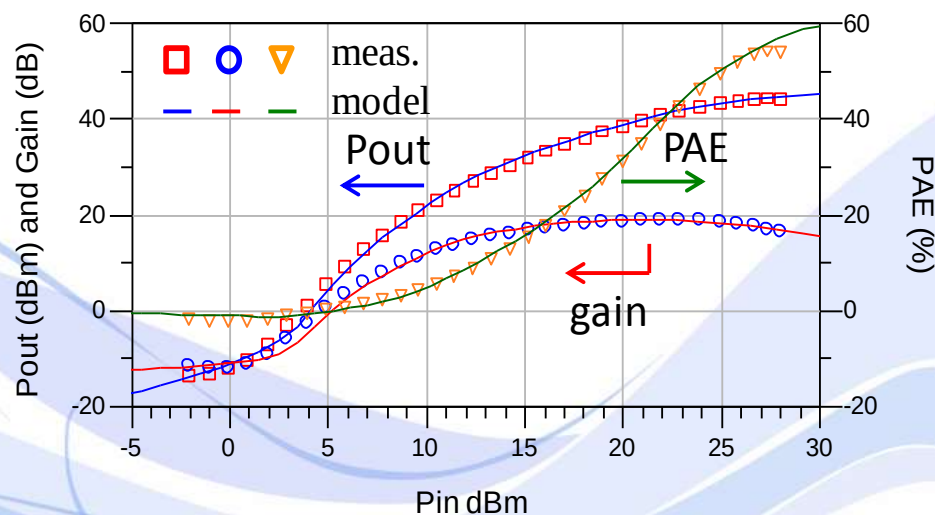
Model validation

- Large-signal

Model validation of a 8x75 μm GaN HEMT with load-pull measurements performed at 6 GHz for optimum PAE load impedance in class-AB



Model validation of a 8x400 μm GaN HEMT with load-pull measurements performed at 3 GHz for the optimum Pout load impedance in class-B

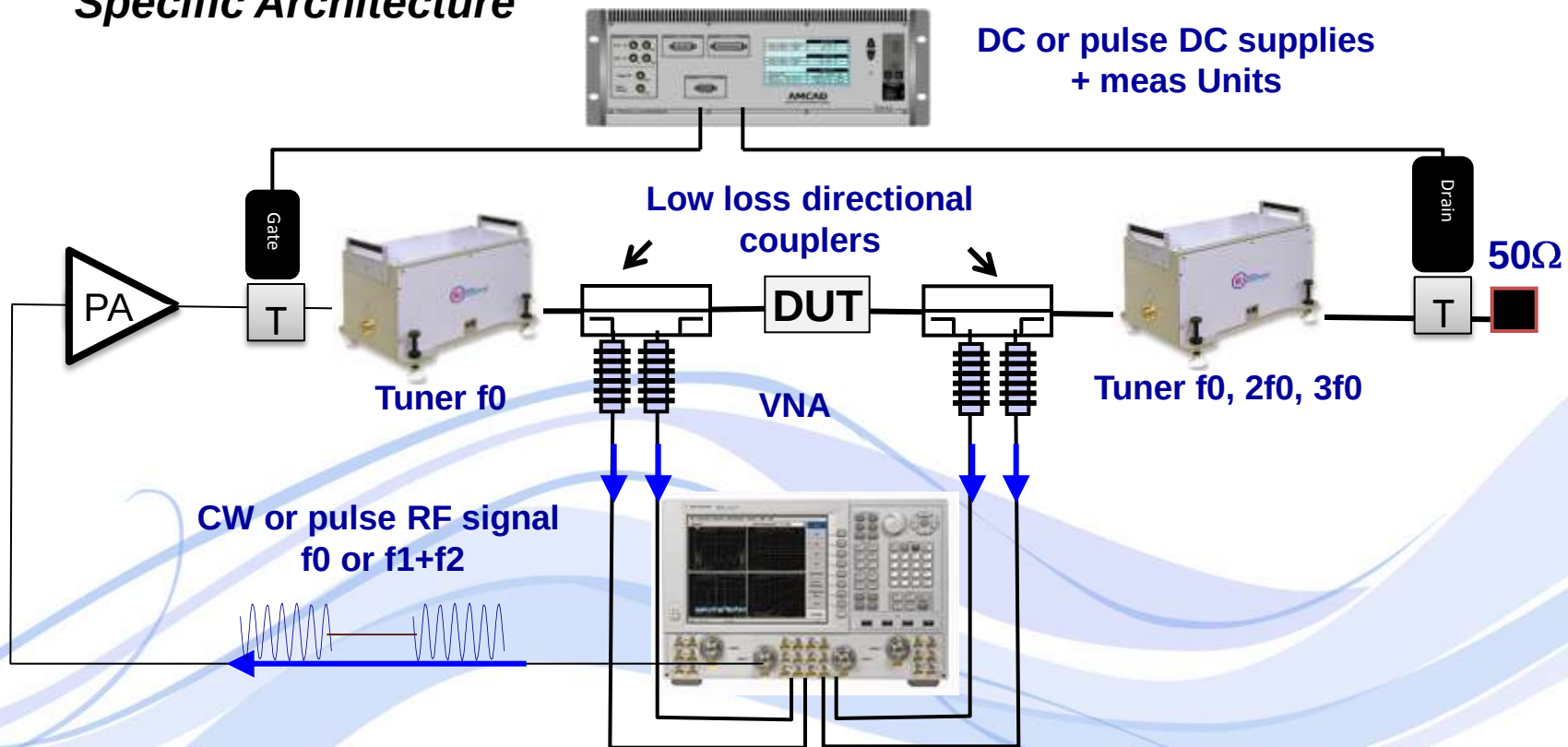




Model validation

VNA Based load pull system is preferred for model validation

Specific Architecture

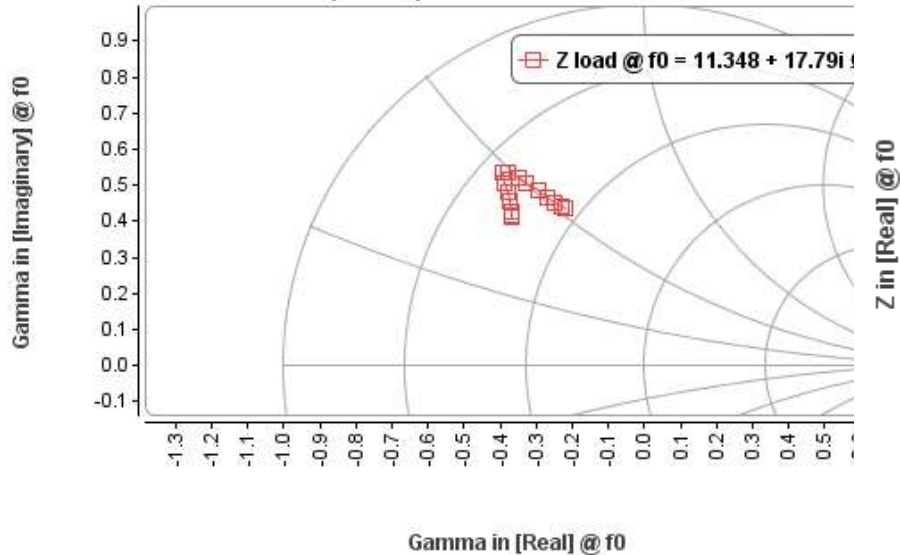




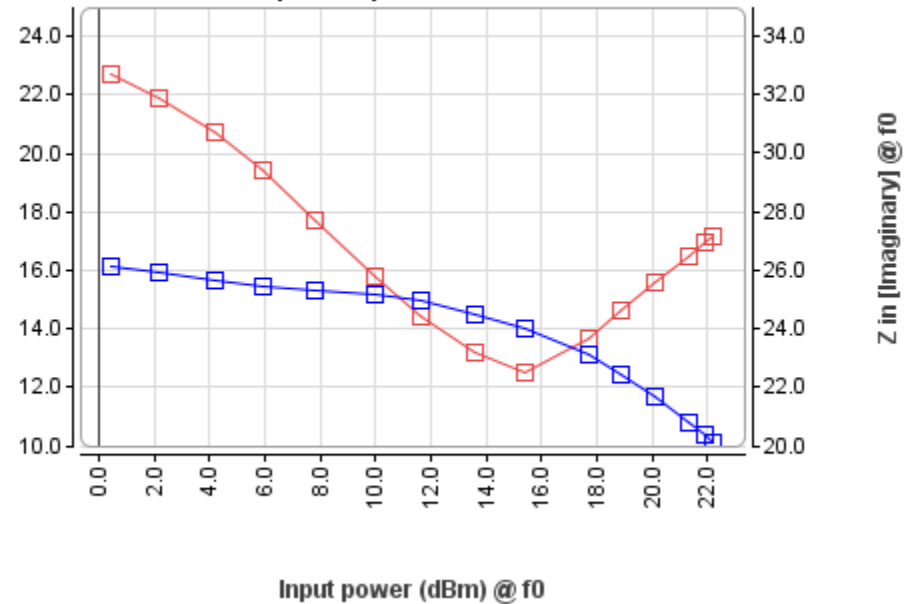
Model validation

VNA Based load pull system is preferred for model validation

Transistor input impedance vs Power level



Transistor input impedance vs Power Level



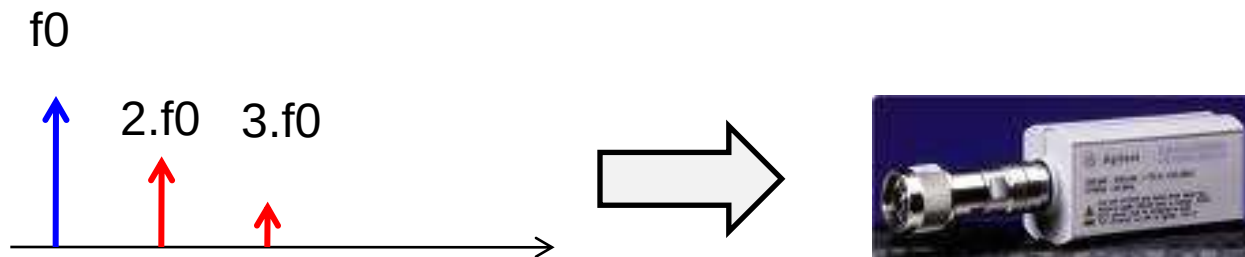


Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Power meter based system are wideband measurement system



VNA based system can be narrowband measurement system



More information
for model
validation or
efficient design

Your complete measurement & modeling solutions partner!



VNA based Load Pull systems

Some Measurement definition

P_{in}: Power delivered to the DUT by the source

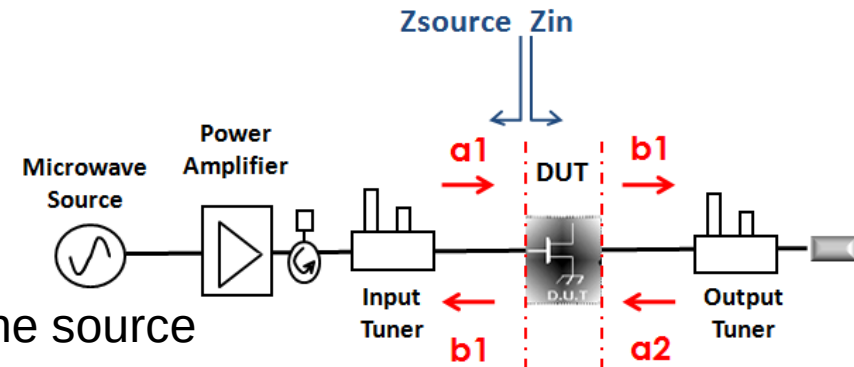
$$P_{in} = \frac{1}{2} (|a1|^2 - |b1|^2) = \frac{1}{2} |a1|^2 * (1 - |\Gamma_{in}|^2)$$

P_{out}: Power delivered to the load impedance

$$P_{out} = \frac{1}{2} (|b2|^2 - |a2|^2) = \frac{1}{2} |b2|^2 * (1 - |\Gamma_{load}|^2)$$

Power gain is the ratio of the power delivered to the load (**P_{out}**) to the power delivered to the transistor by the source (**P_{in}**).

$$Gain_{power} = \frac{P_{out}}{P_{in}} = \frac{|b2|^2(1 - |\Gamma_{load}|^2)}{|a1|^2(1 - |\Gamma_{in}|^2)}$$

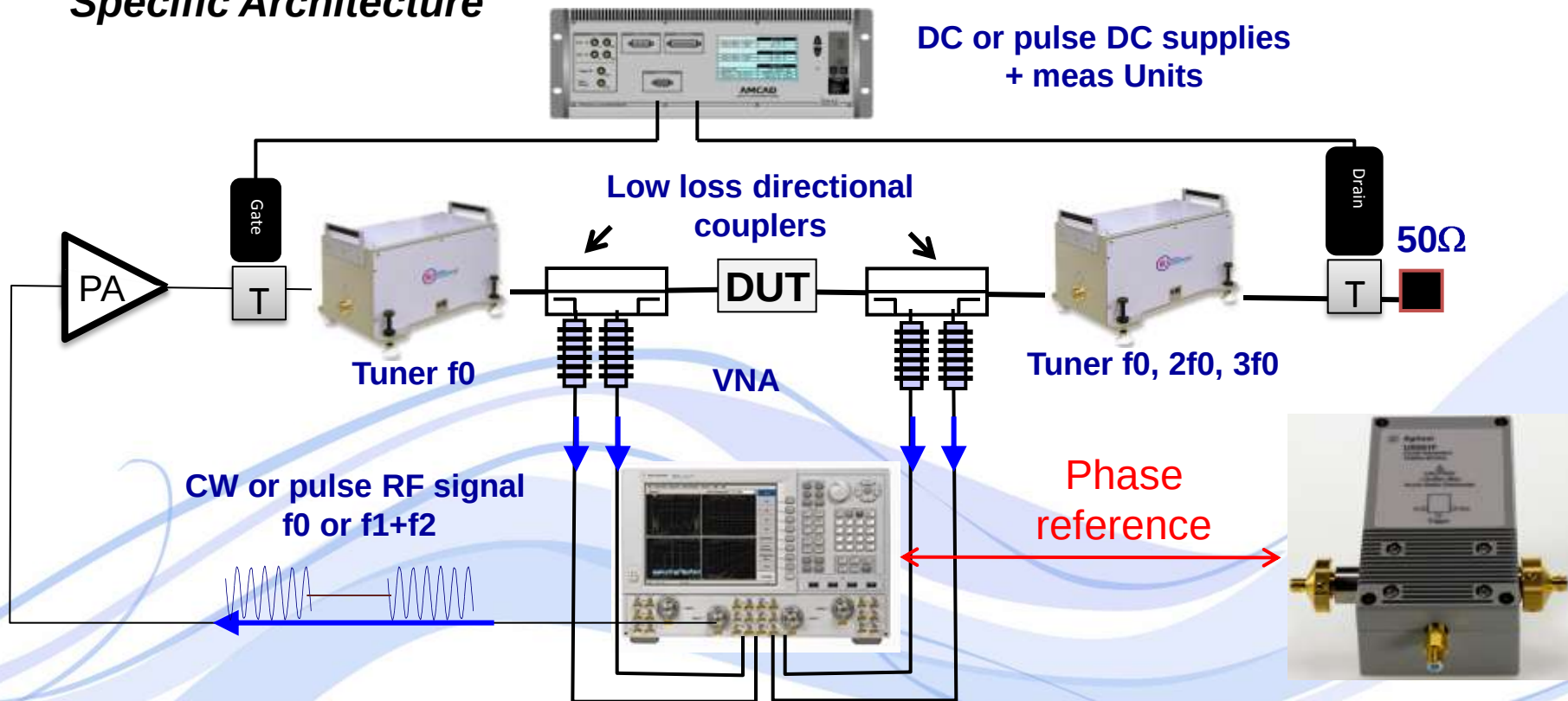




Model validation

VNA Based load pull system is preferred for model validation

Specific Architecture



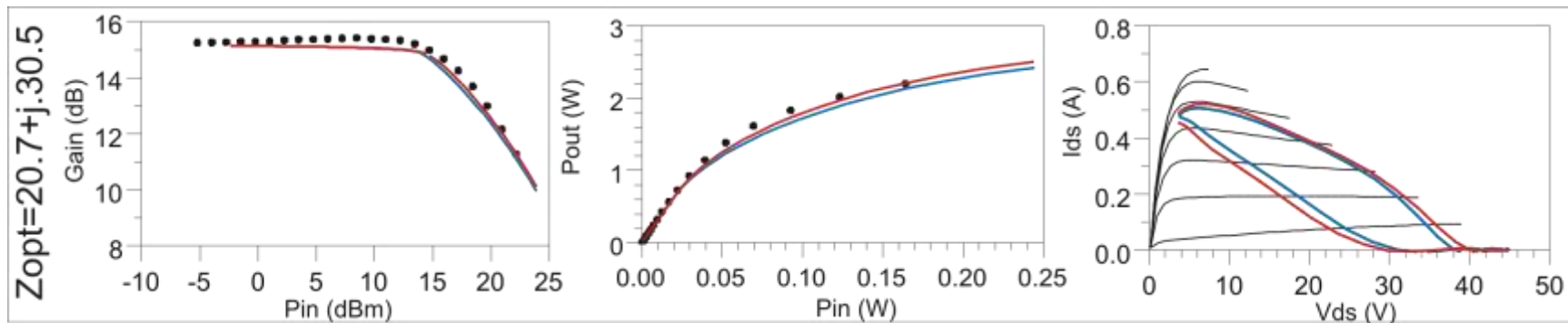


Agilent Technologies

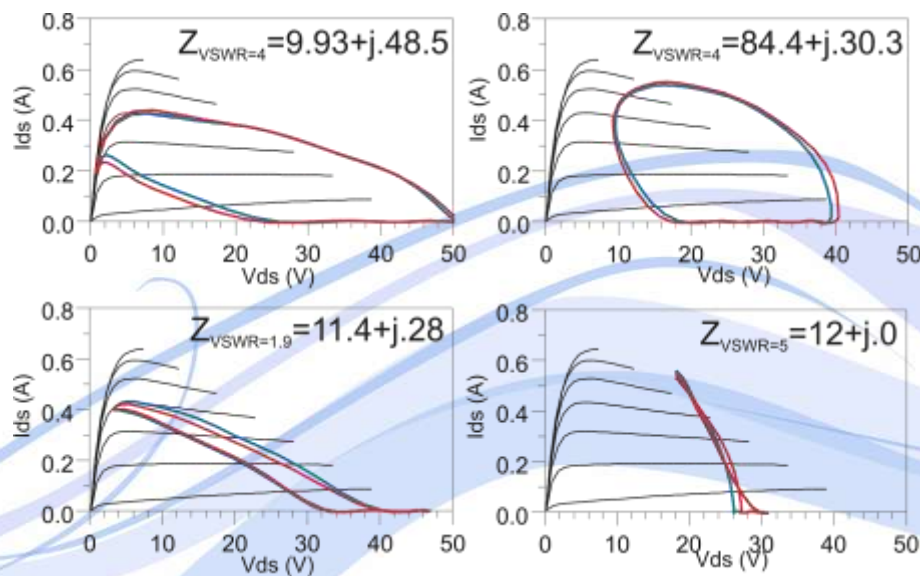


AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Large signal impact - class AB, 25V, 10 GHz – Comparison with measurements



With non optimal loads :



→ Time domain load pull measurements

→ Deembedding in the intrinsic reference plane

→ Parasitic extrinsic elements must be accurately extracted by previous S parameter measurements



Testing Model Validity with Pulsed IV Simulations

(Using Agilent ADS)

- Is your FET model suitable for high-frequency design?
 - Is dynamic behavior included?
 - Is it well-fitted for high-frequency, large signals?
 - What is the model's valid range of use?

“Accounting for Dynamic Behavior in FET Device Models,” Microwave Journal, July, 2011: <http://cp.literature.agilent.com/litweb/pdf/5990-8706EN.pdf>

Download the Pulsed IV Curve Simulation DesignGuide for FETs:
<http://edocs.soco.agilent.com/display/eesofkcads/Pulsed+IV+Curve+Simulation+DesignGuide+for+FETs>

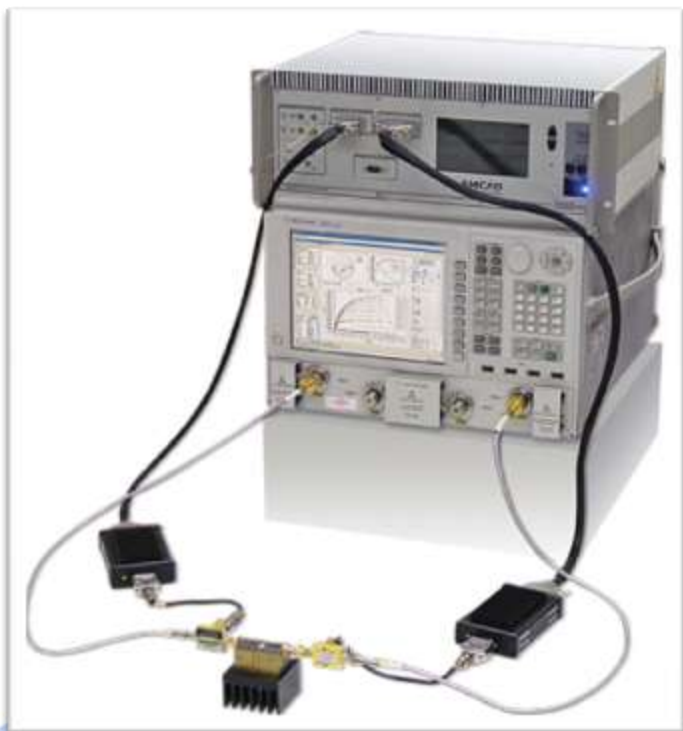


Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Maury – Agilent – AMCAD Solution



Your complete measurement & modeling solutions partner!



Technical Feature

Compact Transistor Models: The Roadmap to First-Pass Amplifier Design Success

Amplifier designers have been making use of modern transistor models since their first appearance in the mid-1970s. Models have allowed engineers to create advanced designs with first-pass success, without the need for multiple prototypes and design iterations. But with so many different modeling techniques, how does one select which one to use? The three most common types of models used in industry today are: physical models, compact models and behavioral models.

Physical models, as their name suggests, are based on the physics of the device technology. These models are dedicated to the transistor itself and not the overall circuit. Due to the nature of the model, complex model equations have to be used, which can lead to time-consuming simulations. The advantage of the physical model is that it can be successfully used over the largest operating range, compared to alternate methods, since equations are used to describe complex physical rules rather than actual measurement results.

Compact transistor models, based on measured IV and S-parameters, allow designers to shift focus from transistor design to circuit design. Extracted from quasi-isothermal pulsed IV and pulsed S-parameter data and validated with load-pull characterization, compact transistor models contain a reduced set of parameters. Unlike other model types, compact models take into account complex phenomena, such as electro-thermal and trapping effects. For simulations under nonlinear operating conditions, responses to complex modulated signals (such as EVM or ACPR) are accurately predicted as low-frequency and high-frequency memory effects are taken into account. Compact transistor models are ideal for die-level applications, as developing such a model from IV and S-parameters is straightforward and relatively quick. Packaged-transistor models need to include a die-level model as well as a bonding model and package model, and consequently can be time consuming and costly.

Behavioral models, based on frequency domain measurements, are far less flexible than physical or compact transistor models, but can be easily developed for any type of component (including die-level or packaged transistors). Behavioral models are considered “black-box” models, where only the response of the component to circuit design. Extracted from quasi-isothermal pulsed IV and pulsed S-parameter data and validated with load-pull characterization, compact transistor models contain a reduced set of parameters. Unlike other model types, compact models take into account complex phenomena, such as electro-thermal and trapping effects. For simulations under nonlinear operating conditions, responses to complex modulated signals (such as EVM or ACPR) are accurately predicted as low-frequency and high-frequency memory effects are taken into account. Compact transistor models are ideal for die-level applications, as developing such a model from IV and S-parameters is straightforward and relatively quick. Packaged-transistor models need to include a die-level model as well as a bonding model and package model, and consequently can be time consuming and costly.

Behavioral models, based on frequency domain measurements, are far less flexible than physical or compact transistor models, but can be easily developed for any type of component (including die-level or packaged transistors). Behavioral models are considered “black-box” models, where only the response of the component to circuit design.

TONY GASSELING
Amcad-Engineering, Limoges, France



Fig. 1 Type of large-signal models

Microwave Journal – March 2012

Technical Feature

“Compact Transistor Models: The Roadmap to First-Pass Amplifier Design Success”



Agilent Technologies



AMCAD Engineering
Advanced Modeling for Computer-Aided Design



Hiroyuki Maehara

Hiro Maehara
Applications Expert – Agilent Technologies
hiroyuki_maehara@agilent.com



Tony Gasseling

Tony Gasseling
General Manager– AMCAD Engineering
gasseling@amcad-engineering.fr



Steve Dudkiewicz

Steve Dudkiewicz
Director, Business Development – Maury Microwave
sdudkiewicz@maurymw.com

Your complete measurement & modeling solutions partner!